Disentangling Ambiguity and Asymmetry in the Attraction Effect: When Comparative Processing Drives Context Dependence

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Abstract

The attraction effect—an increased preference for a target when an inferior decoy is added—has been widely replicated yet remains strikingly fragile across seemingly minor changes in stimulus construction and task demands. A common intuition is that difficult inter-attribute trade-offs make people lean more on dominance, but this conflates two distinct ingredients: ambiguity between the core options (Target vs. Competitor) and asymmetric ease of decoy-related comparisons. Across three experiments, we disentangle these components and show that attraction arises when—and only when—decoy-related comparisons are selectively easier once core-option ambiguity is in place. Experiment 1 establishes robust attraction in a mixed perceptual–numerical choice task (circle area + numerical price), confirming that appropriately constructed perceptual stimuli produce positive asymmetric dominance. Experiment 2 uses an adjustment paradigm to measure inter-attribute "equivalence regions" and demonstrates that the same trade-off difficulty that supports attraction also induces broad competitor-referenced regions that often encompass both target and decoy, especially for circle–price stimuli relative to the bar stimuli. Experiment 3 then holds Target–Competitor difficulty approximately constant while manipulating the relative ease of decoy comparisons with controlled fraction triplets, revealing that attraction and its response-time signature emerge precisely when decoy-related comparative asymmetry is high. These results locate the source of the attraction effect in asymmetric comparative processing rather than structural dominance alone, reconciling its robustness in principle with its empirical elusiveness in practice.

Keywords: Preference reversals, attraction effect, asymmetric dominance

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General Introduction

Consider choosing between two magazine subscriptions: digital-only access for $59, or digital plus print access for $125. Many people may be torn between the lower price of the digital-only option and the added value of receiving both formats. Now suppose a third option is introduced: print-only access for $125. Although this new option is clearly inferior to the digital-plus-print option, its presence can make the latter appear more attractive.

This phenomenon is known as the *attraction effect*, or *asymmetric dominance effect*: adding an inferior decoy increases preference for the option that dominates it (Huber et al., 1982). As one of the clearest demonstrations of context-dependent preference, it violates both the Independence of Irrelevant Alternatives and the Regularity Axiom (R. D. Luce, 1977), and has been documented across consumer choice (Huber et al., 1982; Noguchi & Stewart, 2014), perceptual decision-making (Trueblood et al., 2013) , choices under risk(Farmer et al., 2017), and even non-human species (Bateson et al., 2003; Latty & Beekman, 2011; Lea & Ryan, 2015; Parrish et al., 2015; Scarpi, 2011; Shafir et al., 2002). Yet it is also one of the most empirically elusive (Spektor et al., 2021): modest changes in stimulus construction (Rath et al., 2025), presentation format (Spektor et al., 2018, 2022), or task demands can eliminate or even reverse the effect, and well-powered replications have reported effects of precisely zero (Frederick et al., 2014; Trendl et al., 2021). The field is left with an uncomfortable question: why does a seemingly robust phenomenon depend so sensitively on factors that, in principle, should be irrelevant?

One prominent answer focuses on the difficulty of inter-attribute trade-offs. Related work argues that attraction is especially likely when decision makers are unsure whether an advantage on one attribute is worth a disadvantage on another; as Huber et al. (2014) put it, “for the attraction effect to occur, the decision maker should also be unsure whether a ten-point difference in restaurant ratings is worth a $10 price difference” (see also Hedgcock & Rao, 2009; M. F. Luce et al., 2001). In such cases, the decoy can provide a local comparative advantage for one option over the other, making the attraction effect more likely to emerge (Simonson & Tversky, 1992; Tversky & Simonson, 1993).

This idea is closely related to attribute commensurability. When attributes are readily commensurable, or are expressed on a common scale that facilitates direct comparison, decision makers can more easily evaluate options holistically, reducing reliance on context-sensitive attribute-by-attribute comparisons (Banerjee et al., 2024; Hayes et al., 2025). Conversely, when attributes lack a readily available common evaluative currency, the exchange rate between them becomes harder to determine, and inter-attribute trade-offs become more difficult to resolve (Chang, 2013; Walasek & Brown, 2024). In this sense, incommensurability is not identical to trade-off difficulty, but it is one of its most important structural sources.

The trade-off difficulty account is correct as far as it goes, but it remains incomplete in an important way. Trade-off difficulty is a property of the attribute space, not of any single pairwise comparison. When attributes are difficult to trade-off/integrate, the resulting uncertainty need not be confined to the comparisons between the target and competitor; it can also extend to comparisons involving the decoy. In particular, the same manipulation that creates ambiguity between the core options can also make the competitor–decoy comparison difficult to resolve, while leaving the target–decoy comparison comparatively more decisive under a standard asymmetric-dominance construction. Thus, what appears to be a single explanation in terms of difficult trade-offs may in fact bundle together two distinct ingredients: ambiguity between the core options (Huber et al., 2014) and comparative asymmetry in decoy processing (Huber et al., 1982; Rath et al., 2025).

Trade-off difficulty therefore helps explain when attraction is likely to arise, but because it can induce both ingredients simultaneously, it does not by itself tell us which ingredient is causally responsible.

To state this more precisely, the attraction effect depends on two distinct ingredients. The first is ambiguity between the core options, that is, whether the target and competitor are sufficiently close that the decision maker lacks a strong prior preference between them (Huber et al., 2014). We refer to this as core-option ambiguity, denoted A. The second is comparative asymmetry in decoy processing, that is, whether the target–decoy comparison is selectively easier or more decisive than the competitor–decoy comparison, allowing dominance to be exploited as a path of least resistance (Huber et al., 1982; Rath et al., 2025). We refer to this as decoy-related comparative asymmetry, denoted S. Trade-off difficulty often affects both ingredients together, but they are conceptually distinct and need not vary in lockstep.

The identification problem follows directly. Because difficult or incommensurate attribute constructions often induce both core-option ambiguity and decoy-related comparative asymmetry at once, whereas easy or highly commensurate constructions tend to induce neither, much of the extant literature has effectively sampled only the diagonal of the (A, S) design space: robust attraction at $(1, 1)$ and null effects at $(0, 0)$ (Rath et al., 2025). Yet these diagonal cases do not distinguish between competing causal architectures. Both an AND account, on which attraction requires both ingredients, and an OR-like account, on which either ingredient is sufficient, are consistent with the same diagonal pattern. Table 1 illustrates this logic and maps the four cells to the present experiments. Distinguishing them requires the critical off-diagonal test: $(1, 0)$, where ambiguity is present $(A = 1)$ but comparative asymmetry is absent $(S = 0)$. If attraction disappears in that case, then comparative asymmetry is the operative causal lever rather than a mere by-product of trade-off difficulty. The $(0, 1)$ cell is theoretically degenerate and untestable: strong core preferences leave no room for

Table 1

Design Space of Core-Option Ambiguity (A) and Decoy-Related Comparative Asymmetry (S)

	S = 0	S = 1
A = 0	Null (A0S0) Exp. 2 bars	*Theoretically degenerate* (A0S1; untestable)
A = 1	Critical test (A1S0; Exp. 3 Low $\triangle$Diff)	Attraction (A1S1; Exp. 1, 2 circles & Exp. 3 High $\triangle$Diff)

Note. Diagonal: A0S0 (prior lit + Exp. 2 bars), A1S1 (Exp. 1+2 circles). Off-diagonal: A1S0 tested (Exp. 3); A0S1 untestable since strong core preference leaves no room for decoy shift.

decoy influence.

The present manuscript implements this logic across three experiments. Experiment 1 establishes that stimulus constructions associated with difficult inter-attribute trade-offs can produce robust attraction in a mixed perceptual–numerical paradigm. Experiment 2 then tests why that result is not yet mechanistically diagnostic: using a preregistered adjustment paradigm, it directly measures the geometry of inter-attribute trade-offs and shows that the same incommensurability that supports attraction can also induce broad regions of subjective equivalence that encompass both target and decoy offsets, thereby confounding ambiguity between the core options with asymmetry in decoy-related comparisons. Experiment 3 removes this confound by holding core-option ambiguity approximately constant while manipulating decoy-related comparative asymmetry through controlled fraction stimuli, thereby implementing the critical $(1, 1)$ versus $(1, 0)$ test. Across these studies, we argue that core-option ambiguity is the prerequisite context in which attraction can arise, but comparative asymmetry in decoy processing is the causal lever that produces the shift itself. This account helps explain why the attraction effect can appear robust in principle yet elusive in practice.

Experiment 1: Mixed Perceptual–Numerical Choice Task

Introduction

A central claim in the attraction-effect literature is that difficult inter-attribute trade-offs promote asymmetric dominance: when attributes resist integration into a common scale, decision makers may be less able to resolve the core trade-off directly and may rely more on comparative heuristics, creating conditions under which a classically defined decoy can influence choice (Hedgcock & Rao, 2009; M. F. Luce et al., 2001). Recent demonstrations of attraction with perceptual stimuli are consistent with this idea (Rath et al., 2025), but the trade-off-difficulty account has rarely been tested directly using a novel stimulus class designed to instantiate incommensurability in an objective decision task. Experiment 1 provides such a test.

We used a mixed perceptual–numerical paradigm in which each option combined a visual perceptual attribute (circular disc area) with a symbolic numerical attribute (price), and participants selected the option with the lowest price per unit area. This task instantiates attribute incommensurability in a direct sense: estimating a continuous visual quantity and relating it to a symbolic number provides no obvious common evaluative scale. If attraction is promoted by inter-attribute trade-off difficulty, then this stimulus class should produce a reliable attraction effect. Experiment 1 was designed to test that prediction.

Decoys were defined at the attribute level following Huber et al. (1982), and all three canonical decoy types (range, frequency, and range–frequency) were included to assess generality across decoy configurations. However, no independent measure of item-level comparative asymmetry was obtained for these stimuli, unlike in recent pairwise-discrimination approaches developed for purely perceptual stimuli (Rath et al., 2025). Experiment 1 therefore operates within the framework of the prior literature: the task is constructed to instantiate inter-attribute trade-off difficulty, and dominance is defined at the attribute level following the standard decoy logic. Whether any resulting attraction effect genuinely reflects the mechanism implied by the trade-off-difficulty account—or instead reflects a conflation of distinct structural ingredients—is precisely the question that Experiments 2 and 3 are designed to address.

Methods

Participants

Thirty-six participants (mean age = 21.11 years, SD = 3.42; 27 male, 9 female) with normal or corrected-to-normal vision took part in the experiment after providing informed consent. Six participants were excluded for failing catch trials (accuracy < 0.8), leaving a final sample of 30 participants for analysis. All procedures were approved by the relevant institutional ethics committee.

Stimuli

Each trial displayed three black-filled circular discs on a white background, arranged in a triangular configuration around the center of the screen with vertical positions jittered across trials. Figure 2 illustrates an example trial. Each disc represented a hypothetical alloy, with its price displayed as a numerical value at the center of the disc in arbitrary units.

The decision-relevant attributes were the perceptual area of the disc and its numerical price. Participants were instructed to choose the disc with the lowest price per unit area, requiring integration of a perceptual attribute (area) with a symbolic numerical attribute (price).

Stimuli were generated following procedures adapted from Trueblood et al. (2013). One set of discs was sampled from a bivariate normal distribution with a mean area of 28,000 pixels and a mean price of 28 units. The variances for area and price were 28,000 pixels and 28 units, respectively, with no correlation between dimensions. A second set of discs was matched in price per unit area but had twice the area of the first set. These two sets served as the target and competitor, counterbalanced across trials.

Decoy discs were generated such that, in the attribute space, they were

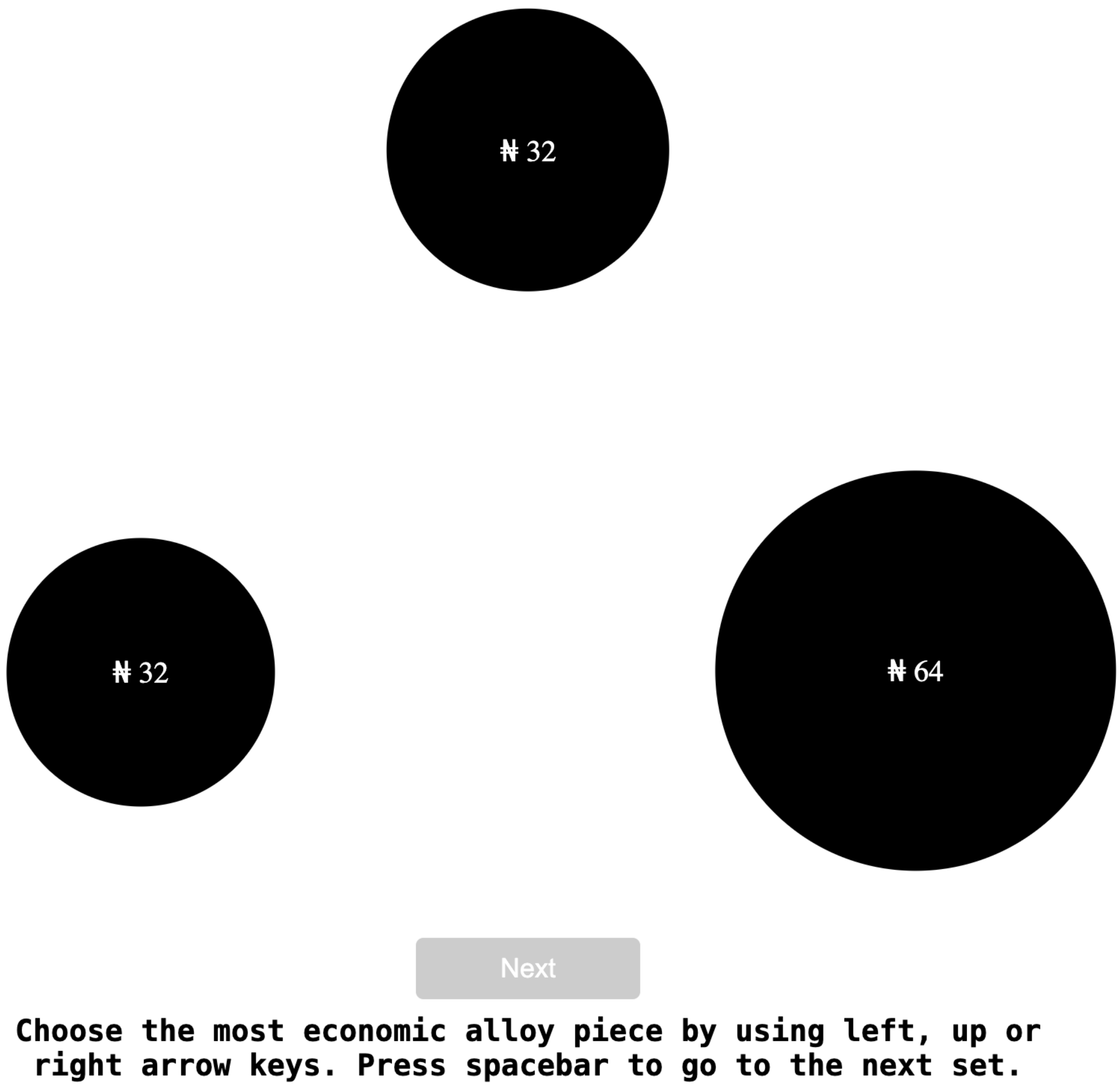

Figure 1

Example Trial in Experiment 1

positioned closer to one of the two core options, creating two contextual conditions. All three standard decoy types—range, frequency, and range-frequency decoys—were included (Huber et al., 1982).

Design and Procedure

Participants completed 36 trials in each contextual condition, resulting in 72 experimental trials. In addition, 12 catch trials were included, in which one option clearly had the lowest price per unit area. The full set of 84 trials was presented in randomized order.

On each trial, participants selected one of the three discs using the left, up, or right arrow keys. They were instructed to respond as quickly and accurately as possible. No time limits were imposed.

Measures

Choice frequencies for the target, competitor, and decoy were recorded on each trial. Context effects were quantified using the equal-weights version of the Relative Choice Share of the Target (RST_{EW}) (Katsimpokis et al., 2022). Response times were recorded for exploratory analyses.

Results

Context effects were quantified using the equal-weights version of the Relative Choice Share of the Target (RST_{EW}; (Katsimpokis et al., 2022)). RST_{EW} captures the relative frequency with which the target is chosen over the competitor across decoy-favoring contexts, with a value of 0.5 indicating no context effect. RST_{EW} was computed as:

$$RST_{\text{EW}} = \frac{1}{2}\left(\frac{T_X}{T_X + C_X} + \frac{T_Y}{T_Y + C_Y}\right),$$

where T_X and C_X denote the number of target and competitor selections, respectively, when the decoy favored option X, and T_Y and C_Y denote the corresponding counts when the decoy favored option Y.

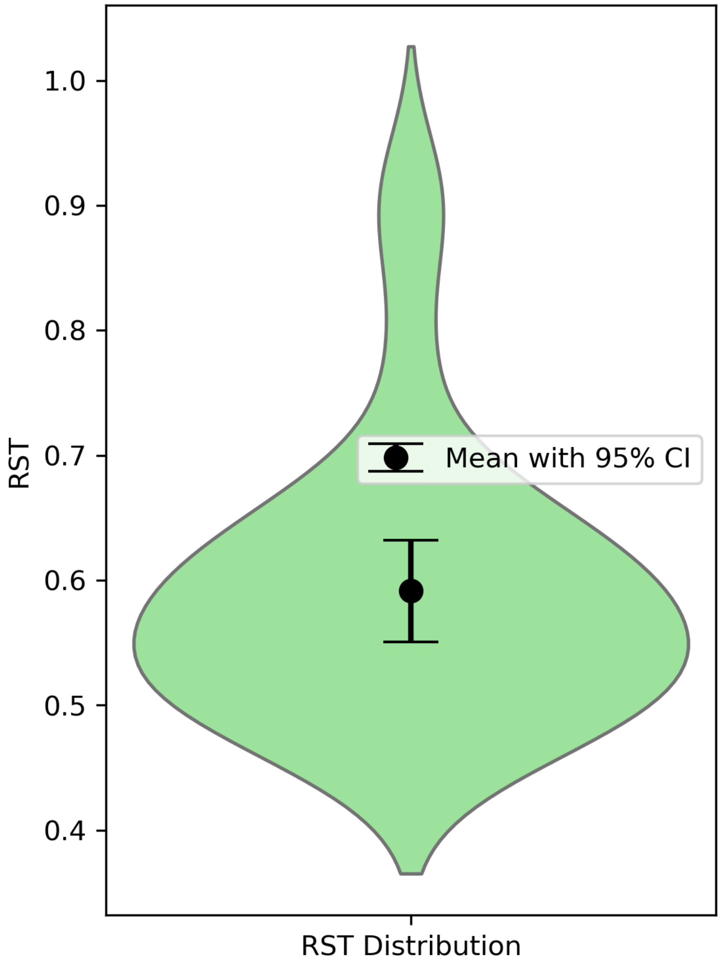

Figure 2

RST Distribution in Experiment 1

Across participants, the mean RST_{EW} was $M = 0.591$ ($SD = 0.109$), exceeding the null value of 0.5. A one-sample one-tailed t-test indicated that this difference was statistically significant, $t(29) = 4.593$, $p < .001$, with a large effect size (Cohen's $d = 0.839$).

A Bayesian one-sample t-test further supported this result, yielding strong evidence in favor of the alternative hypothesis that $\text{RST}_{\text{EW}} > 0.5$ (Bayes factor $\text{BF}_{10} = 273.75$, Cauchy prior with $r = 0.4$). Figure 2 displays the distribution of RST_{EW} values across participants.

Response time analyses were conducted for exploratory purposes. Mean response times did not differ systematically across decoy contexts and are therefore not discussed further here.

Discussion

Experiment 1 demonstrated a robust attraction effect in a mixed-attribute decision task involving both perceptual and numerical information. Participants reliably preferred the target over the competitor when an asymmetrically dominated decoy was introduced, even though the decision required integrating a perceptual attribute (area) with a symbolic numerical attribute (price) to compute price per unit area. This finding replicates prior demonstrations of attraction effects in non-preference-based perceptual tasks (Rath et al., 2025; Trueblood et al., 2013) and extends them to settings that combine perceptual and numerical attributes.

Importantly, Experiment 1 was not designed to isolate the source of dominance asymmetry. As in many perceptual and criterion-based attraction studies, the target and competitor were matched on the decision-relevant criterion, ensuring no objective advantage prior to the introduction of the decoy. However, this design choice also leaves open whether the observed attraction effect reflects (i) weak or unstable baseline preference between the target and competitor, (ii) the selective realization of dominance in comparisons involving the decoy, or some combination of the two. In the terms developed in the General Introduction, Experiment 1 establishes attraction in the present paradigm but does not yet separate core-option ambiguity from decoy-related comparative asymmetry.

Thus, while Experiment 1 establishes that attraction effects can arise in mixed perceptual–numerical decision spaces, it does not adjudicate between alternative explanations of how dominance exerts its influence on choice. In particular, the integration of perceptual and numerical attributes may introduce trade-off uncertainty that affects multiple pairwise comparisons simultaneously, making it unclear whether the observed attraction reflects a selective asymmetry in decoy-related comparisons or a broader “region” of indifference that also encompasses the target–competitor trade-off. Experiment 2 therefore directly examines the geometry of inter-attribute trade-offs underlying such decisions, providing a diagnostic test of whether trade-off uncertainty is selective in the manner required to support asymmetric dominance, or instead broad in a way that confounds multiple comparisons.

Experiment 2: Diagnosing Inter-Attribute Trade-Off Geometry

Introduction

Experiment 1 established that the present mixed perceptual–numerical stimulus family can produce a robust attraction effect: when a decoy was introduced, participants reliably preferred the target over the competitor in a circle-area/price task. Yet that result, by itself, does not identify why the decoy shifted choice. Although attraction is often informally attributed to “trade-off difficulty” or attribute incommensurability, such explanations remain mechanistically incomplete unless they specify how difficulty restructures the pairwise comparisons that actually determine choice (Hedgcock & Rao, 2009; M. F. Luce et al., 2001).

The central problem is that difficult inter-attribute trade-offs can alter more than one comparison at once. If a stimulus family makes the target and competitor hard to discriminate, then core-option ambiguity is present, giving the decoy leverage over the choice set (Huber et al., 2014). But if that same difficulty also makes the competitor hard to distinguish from the decoy, then the decoy-related comparison structure is altered at the same time, potentially creating the very comparative asymmetry needed for attraction (Huber et al., 1982; Rath et al., 2025). In other words, inter-attribute trade-off difficulty may not isolate a single ingredient of the effect; it may bundle together core-option ambiguity A and decoy-related comparative asymmetry S.

This bundling creates the key identification problem of the present paper. The broader manuscript argues that attraction depends on two inputs: ambiguity between the core options and selective comparative ease favoring the Target–Decoy relation over the Competitor–Decoy relation. If prior stimulus manipulations tend to produce only the diagonal cases of this space—stimuli with difficult trade-offs yielding $A = 1, S = 1$, and easy or commensurate stimuli yielding $A = 0, S = 0$—then the literature cannot distinguish an AND architecture, in which both are necessary, from an OR architecture, in which either alone would suffice. A positive attraction effect under circle-price stimuli and a null effect under bars (Supplementary section 5, Rath et al., 2025; Spektor et al., 2022) would then be descriptively informative, but still not causally diagnostic.

Experiment 2 was designed to test whether this confound is built into the geometry of the stimulus families themselves. Rather than inferring ambiguity from aggregate choice shares, we directly measured subjective equivalence regions using an adjustment paradigm under the same decision rules used in the choice tasks. This approach allowed us to ask whether the subjective indifference band around the competitor was broad enough to engulf both the jittered target offset and the decoy offset. If so, then the same trade-off geometry that makes the core options ambiguous would also make the Competitor–Decoy comparison difficult, while leaving the Target–Decoy comparison comparatively straightforward, thereby jointly instantiating the $A = 1, S = 1$ corner. Conversely, if the equivalence band were narrow enough to exclude both offsets, then participants should discriminate the competitor from both the shifted target and the decoy, yielding the opposite bundled corner, $A = 0, S = 0$.

To make this test concrete, we compared two stimulus classes with different expected trade-off geometry. We included the circle-price stimuli used in Experiment 1, where participants must integrate a perceptual quantity with a symbolic number, a setting expected to induce broad subjective equivalence regions and thus reproduce the diagonal $A = 1, S = 1$ case. Second, we included near-commensurate bar stimuli (Rath et al., 2025; Spektor et al., 2022), for which the criterion reduces to a simple sum, as a baseline expected to produce narrow equivalence regions and therefore the opposite diagonal case, $A = 0, S = 0$. The inclusion of bars was thus not merely comparative convenience; it was theoretically necessary for showing that the literature's apparent support for attraction under difficult stimuli and null effects under easy stimuli may arise because the two stimulus classes simultaneously toggle both inputs together.

This experiment therefore served a diagnostic, not yet a causal, function in the logic of the paper. Its purpose was to establish whether prior stimulus families indeed confounded ambiguity and comparative asymmetry by coupling them through the width of the subjective indifference region. If that pattern were observed, then Experiment 1's attraction effect would be reinterpreted as consistent with—but not uniquely explained by—the $A = 1, S = 1$ diagonal, and the need for Experiment 3 would become sharp: only by holding Target–Competitor difficulty approximately constant while independently manipulating decoy-related comparative asymmetry could the paper determine whether S is the operative causal lever once A is in place.

Methods

Participants

Participants were recruited from the institutional participant pool and provided informed consent prior to participation. All procedures were approved by the relevant institutional ethics committee.

Data collection followed a sequential Bayesian sampling plan. A minimum of $N = 20$ participants were collected before interim analyses were conducted. After each additional participant, the Bayes factor associated with the primary hypothesis (structural ambiguity difference between stimulus families) was updated. Data collection was terminated when either (a) $\mathrm{BF}_{10} \geq 10$ (evidence for the alternative), (b) $\mathrm{BF}_{01} \geq 10$ (evidence for the null), or (c) the maximum sample size of $N_{\max} = 70$ was reached.

The maximum sample size was determined via conservative power analysis (see Sample Size Determination below). All stopping criteria were specified a priori.

Participants were excluded prior to analysis if they failed more than two catch trials per stimulus block (relative error greater than 20%), did not complete the session, or exhibited clear non-engagement with the adjustment task (e.g., invariant responses across trials). All exclusion criteria were determined a priori.

Sample Size Determination

Sample size planning was based on the primary hypothesis concerning structural ambiguity differences between stimulus families. A conservative frequentist power analysis for the exact McNemar test (two-tailed, $\alpha = .05$, power = .90) assumed that 60% of participants would exhibit circle-only structural ambiguity and 20% would exhibit bar-only structural ambiguity (odds ratio = 3). Under these assumptions, the required fixed sample size was $N = 70$.

Because confirmatory inference was conducted using a sequential Bayesian framework, this value was treated as an upper bound ($N_{\max} = 70$). A Bayesian factor design analysis (simulation-based) was conducted during planning to examine expected stopping behavior under these assumptions.

Design and Procedure

The experiment employed a fully within-subjects design. Each participant completed 80 bar trials and 80 circle trials, along with 20 catch trials (10 per stimulus family). Block order (bar first vs. circle first) was counterbalanced with probability 0.5 across participants. Trials were randomized within blocks using a Fisher–Yates shuffle with a time-stamped random seed to ensure reproducibility. A brief break separated the two stimulus blocks.

On each trial, the reference option was displayed on the left and the adjustable option on the right. The relevant decision rule (sum for bars; ratio for circles) was displayed at the top of the screen. Participants adjusted the slider until satisfied that both options were equivalent in overall value and confirmed their response. No time limit was imposed. Reaction time was recorded from trial onset to confirmation using high-resolution timing.

The primary confirmatory hypothesis concerned differences in structural ambiguity between stimulus families. Sequential Bayesian evaluation of this hypothesis was conducted only after full completion of each participant's data. No interim analyses were conducted within participants. Block order and trial order randomization were independent of stopping decisions.

Operational Definition of Inter-Attribute Trade-Off Geometry

Inter-attribute trade-off geometry was operationalized using an adjustment paradigm. On each trial, participants adjusted one attribute of an option until it was judged equivalent in overall value to a fixed reference option.

For each participant and stimulus family, equivalence dispersion was quantified as the standard deviation of relative criterion error (VE_{rel}). An equivalence region was defined as:

$$\text{Equivalence Band} = k \times VE_{rel},$$

where $k = 1.5$.

An offset was classified as included if its magnitude did not exceed this band. The multiplier $k = 1.5$ was selected apriori as a moderate tolerance threshold. Sensitivity analyses conducted on pilot data demonstrated that the primary structural effect remained positive and strongly supported across $k \in [1.0, 2.0]$ (see Supplementary Materials), indicating that the result is not contingent on the specific choice of equivalence threshold.

Measures

For each trial, the final adjusted value, true match value, absolute error (AE), relative error, and reaction time were recorded. Absolute error was defined as

$$\mathrm{AE} = |\text{response} - \text{true match}|.$$

Relative error scaled this deviation by the true match value. Additional quality-control metrics included boundary-hit frequency and invariant responding across trials.

For structural classification, predefined target (7%) and decoy (10%) offsets were evaluated against each participant's equivalence band. Participants were classified as exhibiting joint structural ambiguity when both offsets were included within their equivalence region.

Results

All analyses followed the preregistered plan. The primary confirmatory hypothesis (H2) concerned structural ambiguity differences between stimulus families. The dispersion hypothesis (H1) was secondary. Bayesian and frequentist results are reported together for transparency.

A total of 22 participants completed the study. Two participants were excluded based on the preregistered catch-trial criterion, yielding a final analysis sample of $N = 20$.

H1: Equivalence Dispersion

Equivalence dispersion (VE_{rel}) was computed separately for bar and circle stimuli for each participant. As preregistered, we tested whether dispersion was larger for circle stimuli.

Circle stimuli exhibited greater dispersion than bar stimuli. A paired-samples t-test confirmed this difference, $t(19) = 3.504$, $p = .0024$, Cohen's $d_z = 0.783$, indicating a medium-to-large effect.

The corresponding Bayesian paired-samples t-test (Cauchy prior, $r = 0.707$) yielded $\mathrm{BF}_{10} = 16.972$, indicating strong evidence that dispersion was greater for circle stimuli than for bar stimuli. The posterior mean difference was 0.345.

These results indicate that circle stimuli increase inter-attribute trade-off dispersion relative to bar stimuli.

H2: Structural Ambiguity

For each participant and stimulus family, predefined target (7%) and decoy (10%) offsets were evaluated against the participant's equivalence band ($k = 1.5$). Participants were classified as exhibiting joint structural ambiguity when both offsets fell within their equivalence region.

Circle stimuli produced joint structural ambiguity in 100% of participants, whereas bar stimuli produced joint structural ambiguity in 25.0% of participants. The risk difference was 0.750.

A McNemar test comparing paired classifications indicated $\chi^2(1) = 13.067$, $p = .0003$.

The preregistered Bayesian paired binary analysis yielded $\mathrm{BF}_{10} = 2048.000$, indicating extremely strong evidence that circle stimuli produce greater structural ambiguity than bar stimuli. The discordant pattern was $k = 15$ circle-only classifications and $m = 0$ bar-only classifications.

Continuous Distance Analysis (Robustness Check)

To complement the binary classification, we examined continuous standardized distances of the target and decoy offsets from each participant's equivalence boundary. These analyses yielded qualitatively similar conclusions, indicating that circle stimuli reduced the standardized separation between offsets and equivalence boundaries relative to bar stimuli (see Supplementary Section S2).

Sensitivity to Equivalence Multiplier

Sensitivity analyses varying the equivalence multiplier ($k \in [1.0, 2.0]$) demonstrated that the structural effect remained positive across the full tested range. These results indicate that the primary structural finding does not depend on the specific choice of $k = 1.5$ (see Supplementary Figure S3).

Sequential Evidence Accumulation

Sequential Bayes factor trajectories for the primary hypothesis are shown in Supplementary Figure S4. The evidence for H2 exceeded $\mathrm{BF}_{10} = 100$ within the observed sample size.

Discussion

Experiment 2 directly addressed the identification problem left open by Experiment 1. Rather than inferring the role of inter-attribute trade-off difficulty from aggregate attraction scores, we measured the geometry of subjective equivalence directly. The results showed that circle-price stimuli produced broader equivalence regions than bar stimuli, and that these broader regions frequently encompassed both the shifted target offset and the decoy offset relative to the competitor.

This pattern is important because it shows that the same trade-off geometry can bundle together two ingredients that the present paper aims to distinguish. When the equivalence region around the competitor is broad enough to include the shifted target, core-option ambiguity is present. When that same region also includes the decoy offset, the Competitor–Decoy comparison becomes difficult for the same structural reason. Under those conditions, the stimulus family does not manipulate ambiguity alone; it simultaneously alters the comparative structure of the decoy relations as well.

The theoretical implication is that positive attraction under incommensurate stimuli and null effects under commensurate stimuli are not, by themselves, causally diagnostic. Such findings are fully consistent with the present AND-gate account, in which attraction requires both core-option ambiguity A and decoy-related comparative asymmetry S. But they are also compatible with alternative accounts so long as prior stimulus manipulations move both inputs together along the diagonal of the design space. Experiment 2 therefore clarifies why much of the prior literature, including contrasts between circle-price and bar-like tasks, could establish phenomenological differences without isolating the operative causal lever.

At the same time, the present results should not be read as showing that broad equivalence regions are themselves sufficient to generate attraction. Experiment 2 did not measure attraction directly, nor did it independently manipulate ambiguity and comparative asymmetry. Its contribution is instead diagnostic: it shows that in the stimulus families examined here, difficult inter-attribute trade-offs naturally create the bundled $A = 1, S = 1$ case, whereas easier or near-commensurate trade-offs tend toward the opposite bundled corner, $A = 0, S = 0$. In this sense, the experiment explains why Experiment 1 was informative but not decisive.

This logic sharpens the role of Experiment 3. If difficult stimulus families simultaneously induce ambiguity between the target and competitor and reshape decoy-related comparisons, then the only way to identify the operative causal ingredient is to break that coupling. Experiment 3 was designed for precisely that purpose: to hold Target–Competitor difficulty approximately constant while manipulating decoy-related comparative asymmetry independently, thereby testing whether attraction still depends on S once A is already in place.

Experiment 3: Fraction Choice and Comparative Processing Asymmetry

Introduction

Experiments 1 and 2 established two critical points. First, attraction effects can arise robustly in mixed perceptual–numerical choice settings involving incommensurate attributes. Second, manipulations that increase inter-attribute trade-off difficulty can simultaneously alter multiple pairwise comparisons. In particular, when options fall within broad regions of subjective equivalence, difficulty in comparing the decoy with the competitor may co-occur with difficulty in comparing the target with the competitor. This coupling creates a confound: any observed attraction effect may reflect asymmetric decoy-related comparison processing, increased ambiguity between the core options, or both.

Experiment 3 was designed to resolve this confound by isolating decoy-related comparative asymmetry while holding Target–Competitor comparison difficulty approximately constant. To achieve this, we moved to a numerical domain using fraction stimuli, which afford precise control over pairwise relations among options. This allowed us to manipulate the relative ease of Target–Decoy and Competitor–Decoy comparisons without materially changing the comparability of the Target and Competitor. In the logic of the present framework, this provides the critical off-diagonal test by holding ambiguity in the core comparison in place while varying comparative asymmetry.

This design permits a direct test of the central hypothesis of the present work: attraction arises when, and only when, decoy-related comparisons are selectively easier or more decisive once core-option ambiguity is present. By controlling Target–Competitor comparison difficulty and systematically varying decoy-related asymmetry, Experiment 3 tests whether asymmetric dominance influences choice through selective decoy elimination rather than through global trade-off difficulty. In addition, trial-level response-time analyses allow us to distinguish the role of deliberation in eliminating the decoy from its role in shaping relative preference between the target and the competitor.

Method

Participants

Fifty-five participants (age range XX–XX years; XX female) were recruited from an online participant pool. All participants provided informed consent prior to participation and were compensated according to institutional guidelines. The study was approved by the institutional ethics committee. Data from all participants were included in the analyses reported below.

Stimuli

Each trial consisted of a choice among three options: a *Target* (T), a *Competitor* (C), and a *Decoy* (D). Each option was defined by two numerical attributes, represented as fractions. Fractions were chosen to allow precise control over pairwise relationships among options while maintaining a common metric across attributes.

The decoy was constructed to be asymmetrically dominated by the target but not by the competitor. Critically, stimulus construction allowed selective manipulation of the relative ease of Target–Decoy and Competitor–Decoy comparisons while holding Target–Competitor comparison difficulty approximately constant. This manipulation formed the basis of the ΔDiff condition described below.

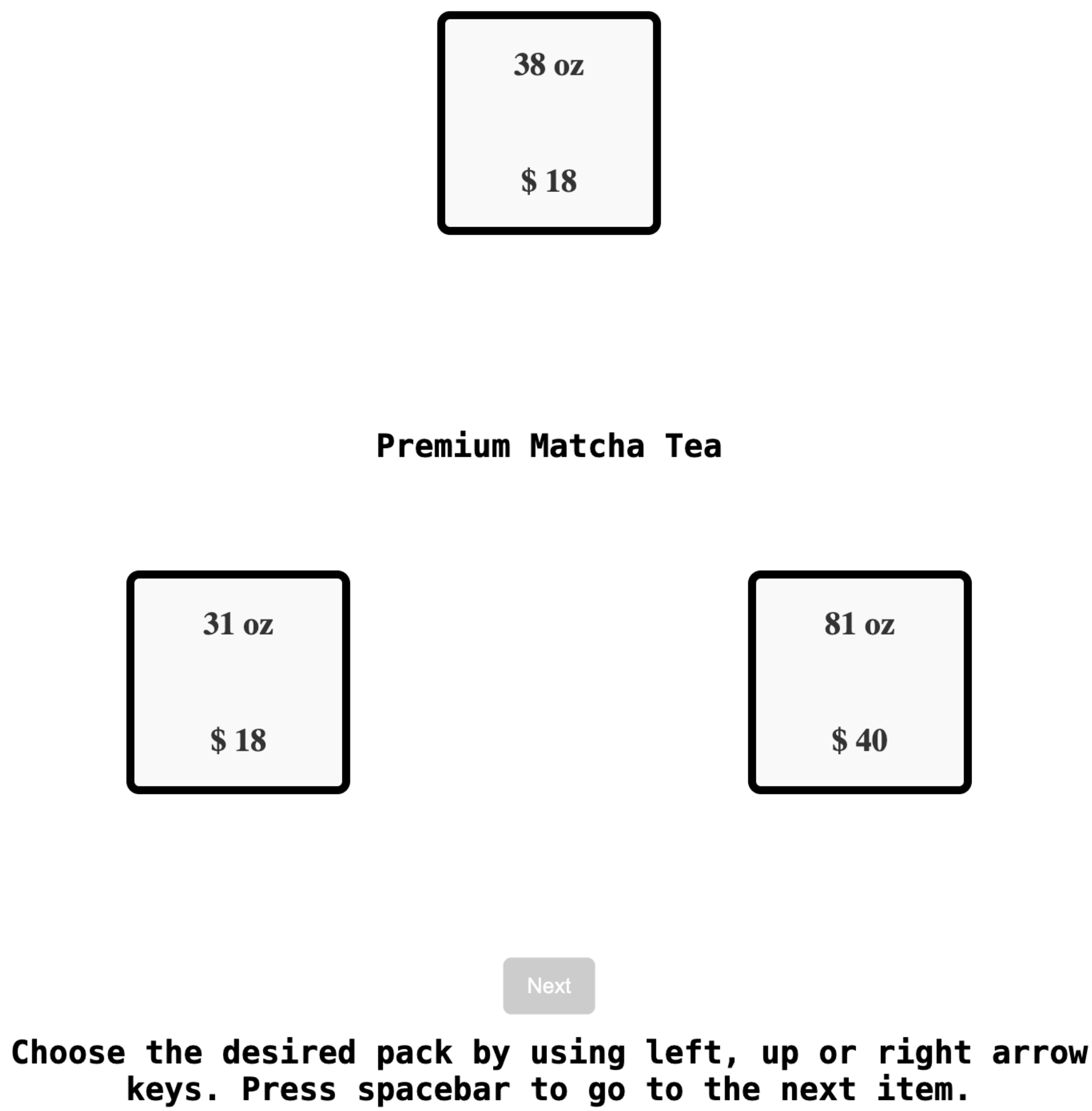

Figure 3

Example Trial for Block H in Experiment 3

Figure 5 shows RST as a function of response-time quartile separately for each ΔDiff and block-order condition. Although RST sometimes appeared to increase with deliberation time, this trend was inconsistent across conditions, motivating a decomposition of choice into decoy elimination and target–competitor preference.

Operational Definition of Comparative Ease

To operationalize the relative ease of pairwise comparisons among options, we quantified comparative discriminability using a normalized geometric-mean distance measure across attributes. For any two options i and j, each defined by attributes a_1 and a_2, comparative ease was defined as:

$$\text{Ease}(i,j) = \sqrt{\left|\log\left(\frac{a_{1i}}{a_{1j}}\right)\right| \cdot \left|\log\left(\frac{a_{2i}}{a_{2j}}\right)\right|}$$

This measure captures joint proportional separation between two options across both attributes while remaining invariant to scale transformations. Larger values correspond to greater separability and, consequently, easier pairwise comparison.

For each trial, three pairwise ease values were computed: Target–Decoy (TD), Competitor–Decoy (CD), and Target–Competitor (TC). The ΔDiff manipulation was defined as the relative difference between TD and CD ease:

$$\Delta\text{Diff} = \text{Ease(TD)} - \text{Ease(CD)}$$

In the High ΔDiff condition, TD comparisons were substantially easier than CD comparisons. In the Low ΔDiff condition, this asymmetry was reduced, rendering TD and CD comparisons more similar in ease. Across both conditions, TC ease was held approximately constant by construction.

The geometric-mean formulation and associated constraints were used solely to guide stimulus selection and ensure controlled manipulation of pairwise comparative asymmetry. These heuristics were used solely to constrain stimulus selection and do not constitute claims about the strategies participants employed during the task. Full details of the heuristic criteria, weighting schemes, sampling procedure, and threshold values are provided in the Supplementary Materials.

A separate supplementary pair-rating validation study confirmed that the intended comparative asymmetry was psychologically realized: the CD–TD subjective difficulty gap was larger in High than Low $\triangle$Diff, while target–competitor (TC) difficulty remained stable across conditions (see Supplementary Section S5)

Design and Procedure

The experiment employed a within-subjects design with two primary conditions corresponding to levels of comparative asymmetry: High and Low $\triangle$Diff. Trials were presented in two blocks, one per condition. Block order was counterbalanced across participants (High–Low vs. Low–High). Each block contained an equal number of trials, and the same Target–Competitor pairs were used across conditions, differing only in the construction of the decoy.

On each trial, the three options were displayed simultaneously on the screen in randomized spatial order. Participants were instructed to select the option they preferred by clicking on it. No time limit was imposed. Choice and response time (RT) were recorded on every trial.

Response Time Processing

Response times were measured from stimulus onset to choice selection. Trials with missing responses or non-positive RTs were excluded prior to analysis (less than X% of trials). RTs were log-transformed to reduce skew. For descriptive analyses, trials were additionally grouped into participant-wise RT quartiles.

Analytic Strategy

Analyses proceeded in three stages. First, overall choice proportions and the relative share of the target (RST) were computed to assess attraction effects across conditions. Second, decoy choice was modeled at the trial level using hierarchical logistic regression with log-transformed RT, $\triangle$Diff condition, and block order as predictors, including participant-specific random intercepts and slopes. Third, analyses of target choice were restricted to trials in which participants selected either the target or the competitor, allowing assessment of target–competitor preference conditional on decoy elimination.

All hierarchical models were estimated in a Bayesian framework using weakly informative priors. Posterior inference focused on credible intervals and region-of-practical-equivalence (ROPE) analyses to assess evidence for practically meaningful effects.

Results

Overall Choice Patterns

Across conditions, the fraction task reproduced canonical attraction patterns when comparative asymmetry was high. The relative share of the target exceeded chance levels under high ΔDiff, whereas under low ΔDiff, target and competitor shares were approximately balanced, yielding RST values near 0.5. This pattern was observed across block orders, indicating that the effect was not driven by presentation sequence.

Decoy Elimination and Response Time

To examine the role of deliberation in decoy processing, we modeled decoy choice at the trial level using a hierarchical logistic regression with RT as a predictor. Across all conditions, the probability of choosing the decoy decreased sharply with increasing response time. This effect was robust across ΔDiff and block order and showed substantial consistency across participants.

Figure 6 illustrates the conditional effect of RT on decoy choice probability. Faster responses were associated with frequent decoy selection, whereas slower responses were overwhelmingly dominated by choices between the target and competitor. This finding indicates that decoy elimination is a time-sensitive process that unfolds during deliberation.

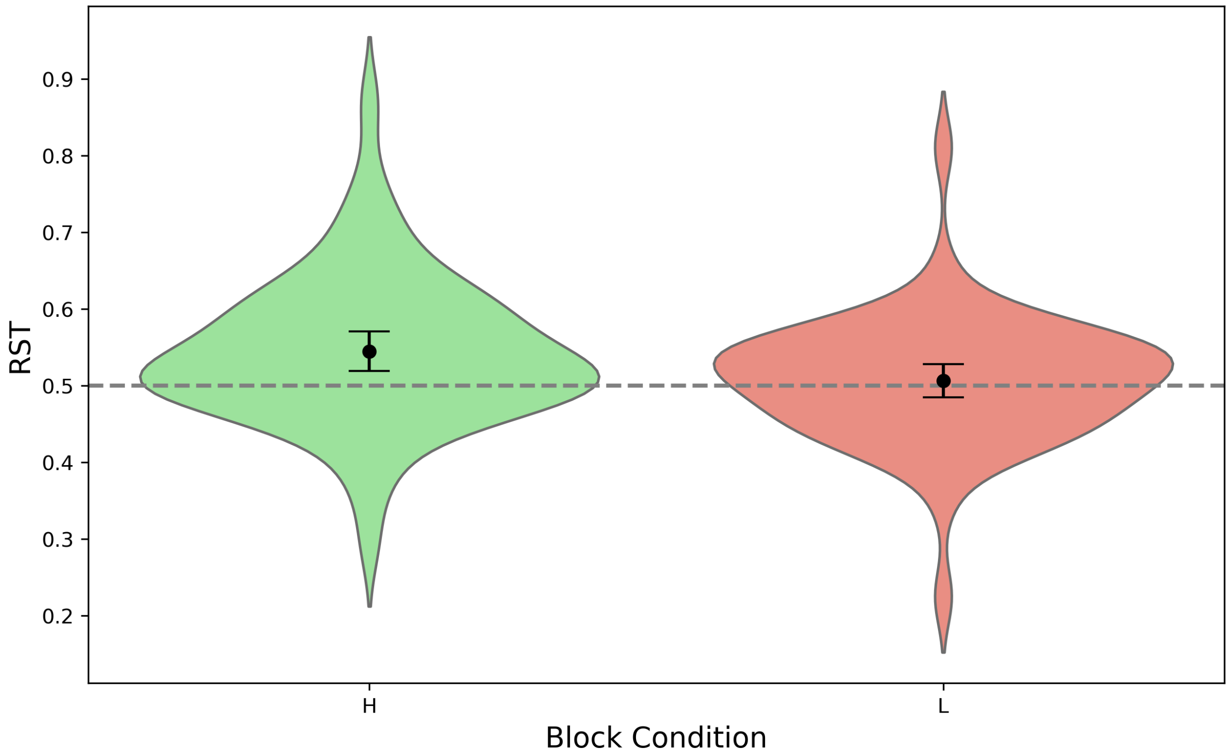

Figure 4

RST Distributions for two blocks in Experiment 3

Target–Competitor Preference Conditional on Decoy Elimination

The critical test of the attraction mechanism concerns relative preference between the target and competitor once the decoy has been eliminated. To assess this, we restricted analyses to trials in which participants chose either the target or the competitor and modeled target choice using a hierarchical logistic regression with RT, ΔDiff, block order, and their interactions as predictors.

Across all conditions, posterior estimates of RT slopes were small, and 95% credible intervals consistently spanned zero. Figure 7 summarizes posterior mean RT effects and credible intervals for each ΔDiff and block-order condition. No condition showed reliable evidence for a systematic increase or decrease in target preference with deliberation time.

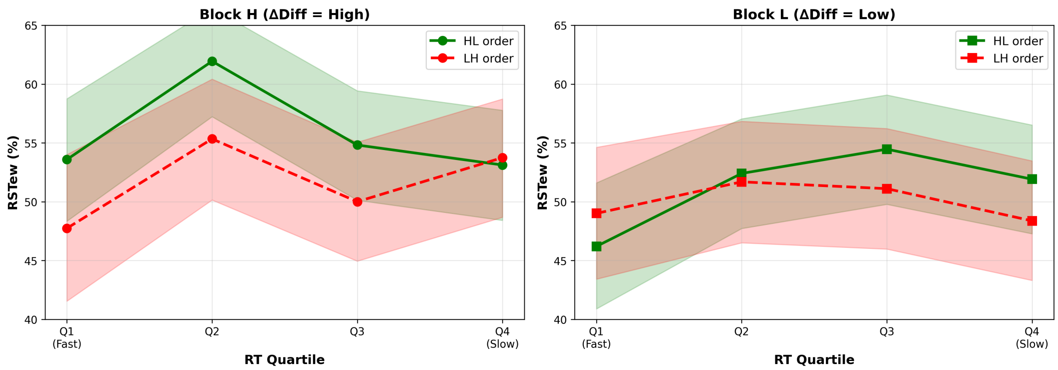

Figure 5

Relative share of the target (RST) across response-time quartiles, plotted separately by △Diff (high vs. low) and block order. Error bars indicate 95% credible intervals.

Region-of-practical-equivalence (ROPE) analyses further supported this conclusion. In several conditions, the majority of posterior mass for the RT slope lay within a narrow ROPE around zero, providing evidence for the absence of practically meaningful RT effects. In remaining conditions, posterior mass was broadly distributed but did not support reliable deviations from zero.

Decomposing Apparent RT Effects

Descriptive analyses of response-time quartiles revealed that target choice frequency often increased with RT, while decoy choice frequency decreased sharply. Figure 8 shows the distribution of target, competitor, and decoy choices across RT quartiles. Increases in target frequency largely mirrored decreases in decoy frequency, whereas target–competitor choice proportions remained comparatively stable across quartiles.

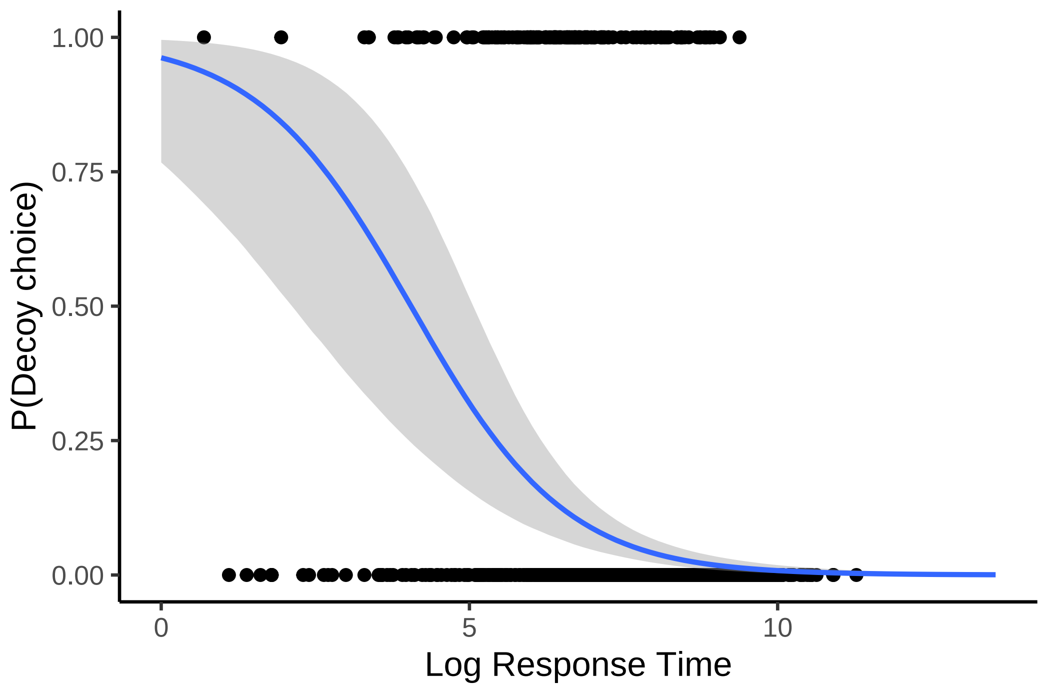

Figure 6

Model-predicted probability of decoy choice as a function of log response time. Shaded regions indicate 95% credible intervals.

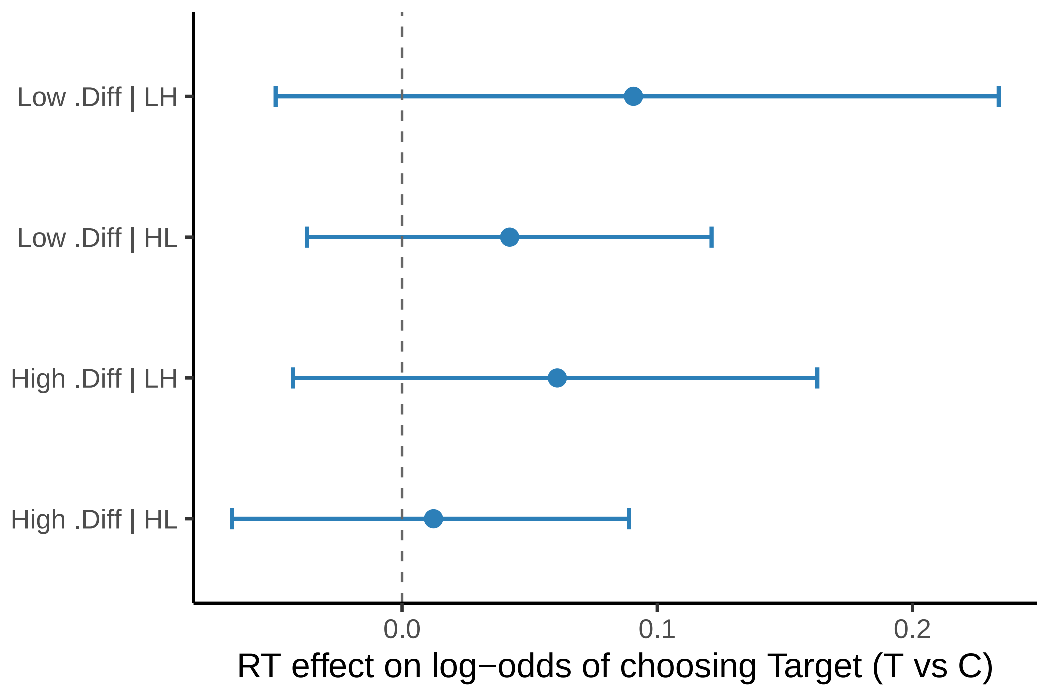

Figure 7

Posterior estimates of the RT slope on target choice (Target vs. Competitor) across ∆Diff and block-order conditions. Points denote posterior means; error bars indicate 95% credible intervals. The dashed line marks zero (no RT dependence).

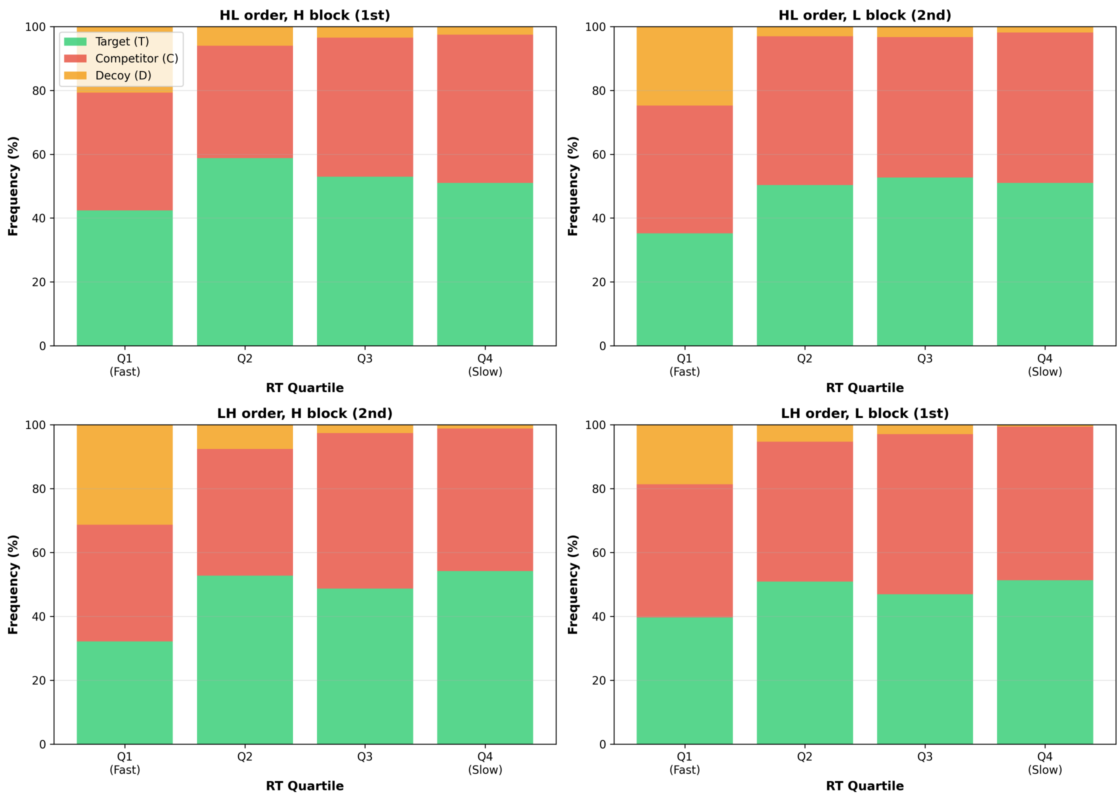

Figure 8

Choice proportions for Target (T), Competitor (C), and Decoy (D) across response-time quartiles. Decoy choice decreases monotonically with RT, whereas Target and Competitor choices increase symmetrically.

Discussion

Experiment 3 yielded three converging findings. First, attraction emerged reliably when comparative asymmetry was high, as indexed by the $\triangle$Diff manipulation, and was attenuated when that asymmetry was reduced. Second, deliberation time strongly predicted decoy elimination across conditions. Third, conditional on decoy elimination, target–competitor preference showed little evidence of systematic modulation by either response time or comparative asymmetry. Taken together, these results indicate that attraction in fraction choice is driven primarily by asymmetric comparative processing involving the decoy, rather than by a gradual strengthening of target preference during extended deliberation.

This pattern is theoretically important because it resolves the confound identified in Experiment 2. There, trade-off difficulty altered the geometry of subjective equivalence in a way that could jointly affect both the core Target–Competitor comparison and decoy-related comparisons. By contrast, Experiment 3 held Target–Competitor comparison difficulty approximately constant while selectively varying decoy-related comparative asymmetry. The resulting shift in attraction therefore identifies comparative asymmetry, rather than global trade-off difficulty, as the operative causal lever once core-option ambiguity is already in place. This interpretation is consistent with the supplementary pair-rating validation showing selectively increased decoy-related comparative difficulty asymmetry in the High ($\triangle$Diff) condition (Supplementary Section S5).

The response-time analyses further clarify the mechanism. Longer response times strongly reduced the probability of selecting the decoy, consistent with time-sensitive dominance detection. However, once the decoy was no longer chosen, response time did not reliably tilt preference toward the target over the competitor. Apparent increases in target choice with longer deliberation therefore primarily reflected progressive decoy elimination rather than amplification of target preference within the core pair.

This decomposition also offers a more specific interpretation of prior findings linking time pressure to the attenuation of attraction effects (Spektor et al., 2021). The present results suggest that time pressure weakens attraction not because it prevents a slow build-up of target preference, but because it truncates the selective comparative processing required to eliminate the decoy and realize the asymmetry in its favor.

General Discussion

The attraction effect has long occupied a paradoxical position in decision research. On the one hand, the addition of an asymmetrically dominated decoy often shifts preferences toward its dominating target across domains and tasks. On the other hand, the effect frequently proves fragile, disappearing under seemingly modest changes in stimulus construction, task structure, or time constraints. The present work resolves this tension by showing that the decisive factor is not structural dominance alone, nor global trade-off difficulty per se, but whether decoy-related comparisons are selectively easier or more decisive once ambiguity between the core options is in place.

Across three experiments, we dissociated these ingredients. Experiment 1 established robust attraction in a mixed perceptual–numerical paradigm. Experiment 2 showed that manipulations based on inter-attribute trade-off difficulty can broaden subjective equivalence regions in a way that jointly affects both the core Target–Competitor comparison and decoy-related comparisons, rendering attraction mechanistically ambiguous. Experiment 3 then broke this coupling by holding Target–Competitor difficulty approximately constant while selectively manipulating decoy-related comparative asymmetry. Together, these findings indicate that core-option ambiguity provides the prerequisite context, whereas asymmetric comparative processing involving the decoy is the operative causal lever.

Asymmetry Must Be Process-Realized, Not Merely Structurally Defined

Classical accounts often characterize dominance asymmetry in terms of structural relations within the option set (Bhatia, 2013; Helgadóttir, 2015; Kaptein et al., 2016; Žofák, 2016) or in terms of the aggregate behavioural patterns those relations are expected to produce (Huber et al., 1982; Rath et al., 2025). In that sense, asymmetry is treated as something that can be specified from the stimuli or inferred from choice outcomes without yet committing to the mechanism by which it exerts influence. Our results show that such structural characterizations are not sufficient. What matters is whether the asymmetry is *comparatively realized*—that is, whether the cognitive system can readily detect and exploit the relevant dominance relation during pairwise evaluation.

Experiment 1 established the phenomenon in a mixed perceptual–numerical paradigm, showing that appropriately constructed stimuli can support attraction, while leaving its precise source to be clarified by Experiments 2 and 3. Experiment 2 then showed why explanations based on inter-attribute trade-off difficulty or incommensurability (Chang, 2013; Walasek & Brown, 2024) are not, by themselves, mechanistically diagnostic. When trade-off difficulty induces broad regions of subjective equivalence, it can blur multiple pairwise comparisons at once, encompassing not only the target–competitor comparison but also decoy-related comparisons. In such cases, structural asymmetry may still be definable at the level of the stimulus configuration, and attraction may still be observable at the behavioural level, yet the process by which the effect is produced remains ambiguous.

Experiment 3 resolved this confound by holding target–competitor comparison difficulty approximately constant while selectively manipulating the relative ease of target–decoy and competitor–decoy comparisons. Under these conditions, attraction emerged precisely when decoy-related comparative asymmetry was high and was attenuated when that asymmetry was reduced. Dominance therefore influences choice not merely because an asymmetry can be specified in the stimuli or detected in behaviour, but because it is selectively realized in comparative processing.

Decoy Elimination and Preference Formation

A second contribution of the present work is the dissociation of decoy elimination from target–competitor preference formation. Across conditions, response time robustly predicted a reduction in decoy choice, consistent with the view that extended processing facilitates the rejection of dominated alternatives. However, conditional on decoy elimination, response time did not reliably shift preference between the target and competitor.

This finding clarifies a recurring ambiguity in the attraction literature. Increases in target choice are often interpreted as evidence of strengthened preference for the target. Our analyses show that such increases can arise purely from the removal of the decoy from consideration, leaving the relative balance between target and competitor unchanged. This decomposition aligns with critiques emphasizing the importance of analyzing full choice distributions rather than aggregated target shares (Katsimpokis et al., 2022; Liew et al., 2016). Further, all reported effects were robust to out-of-sample validation: hierarchical models exhibited stable predictive performance under Pareto-smoothed leave-one-out cross-validation, with no evidence that inferences were driven by influential trials.

Reinterpreting Time Pressure Effects

The present findings also shed new light on the effects of time pressure. Prior work has consistently shown that both external and internal time pressure weaken or eliminate attraction effects (Marini et al., 2020; Spektor et al., 2021). This pattern has often been interpreted as evidence that attraction requires slow, deliberative, compensatory processing.

Our results suggest a more specific interpretation. Time pressure disrupts attraction not because it eliminates comparison altogether, but because it truncates the selective comparative processing that allows decoy-related asymmetries to influence choice. Under limited time, decoy elimination is incomplete, and dominance relations fail to exert their asymmetric influence. This account is consistent with dynamic models of multialternative choice, which predict that context effects emerge only after sufficient comparative evidence has accumulated (Evans et al., 2019; Roe et al., 2001; Usher & McClelland, 2004).

Importantly, this interpretation explains why cognitive load manipulations often fail to eliminate attraction effects (Wedell, 1991; Wedell et al., 2022): generic effort is not the critical factor. Rather, what matters is whether the task affords sufficient time for selective pairwise comparisons to unfold.

Implications for Theories of Context Effects

By locating the source of attraction in asymmetric comparative processing, the present work helps bridge several theoretical traditions. It complements sequential sampling and dynamic accumulation models that emphasize comparison dynamics (Bhatia, 2013; Roe et al., 2001; Trueblood et al., 2014), while also aligning with reason-based and comparison-based accounts of choice (Russo & Dosher, 1983; Simonson, 1989). At the same time, it cautions against explanations that appeal exclusively to attribute trade-off difficulty or incommensurability, because such factors can simultaneously alter core-option ambiguity and the relative ease of decoy-related comparisons.

More broadly, this perspective helps explain why attraction effects appear across diverse species and task domains (Bateson et al., 2003; Parrish et al., 2015), yet remain highly sensitive to seemingly minor design details. Dominance asymmetry is not exhausted by the static structure of the choice set; its influence depends on whether that asymmetry is selectively realized in comparative processing. In this sense, the present findings reconcile the broad scope of the attraction effect with its apparent empirical fragility.

Conclusion

The attraction effect is neither a fragile illusion nor a universal inevitability. It arises when—and only when—core-option ambiguity is in place and comparative processing is asymmetrically structured around the decoy. By disentangling structural dominance from its comparative realization, the present work offers a unifying account of why attraction effects are robust in principle yet elusive in practice. This framework provides a foundation for future work on context effects that moves beyond asking whether attraction occurs toward identifying the precise structural and process conditions under which dominance shapes choice.

Transparency and Data Availability

All data preprocessing steps, analysis code, and experimental materials will be made publicly available on the Open Science Framework upon publication.

Supplementary Methods

S1: Sensitivity analysis of the equivalence band (Exp 2)

The structural ambiguity analysis (H2) defined joint inclusion of the target and decoy offsets relative to each participant's empirical equivalence dispersion (VE_{rel}) using bands of the form $|\Delta| \leq k \cdot VE_{\text{rel}}$. Although the preregistered analysis used $k = 1.5$, we assessed robustness by recomputing gate classifications across $k \in [1.0, 2.0]$ in steps of 0.05, holding the preregistered offsets $\Delta_T = 0.07$ and $\Delta_D = -0.10$. This analysis directly tests whether the structural inclusion of both target and decoy offsets depends on the choice of equivalence threshold.

For each k and stimulus family (bar vs. circle), we computed whether both offsets lay inside the band (joint gate). We report (i) the proportion of participants satisfying the joint gate under circle minus the corresponding proportion under bar, and (ii) a paired-binary Bayes factor BF_{10} from discordant classifications (circle-only vs. bar-only) using the same beta–binomial marginal for H_1 and $\mathrm{Binomial}(p = 0.5)$ under H_0 as in the confirmatory Bayesian pipeline.

Figure 9 summarizes this sensitivity analysis. The proportion difference (circle minus bar; solid blue) and the corresponding scaled $\log_{10}$ Bayes factor, $\log_{10}(\mathrm{BF}_{10})/10$ (solid firebrick), are plotted across values of k, along with reference lines indicating the null effect and the preregistered choice $k = 1.5$.

Across the tested grid, the proportion difference remained strictly positive for $k \in [1.00, 2.00]$. The proportion difference ranged from 0.542 to 0.792, and BF_{10} ranged from 585.14 to 26214.40. These results indicate that the circle–vs.–bar structural contrast is not contingent on the specific choice $k = 1.5$, but instead reflects a robust property of the measured equivalence geometry.

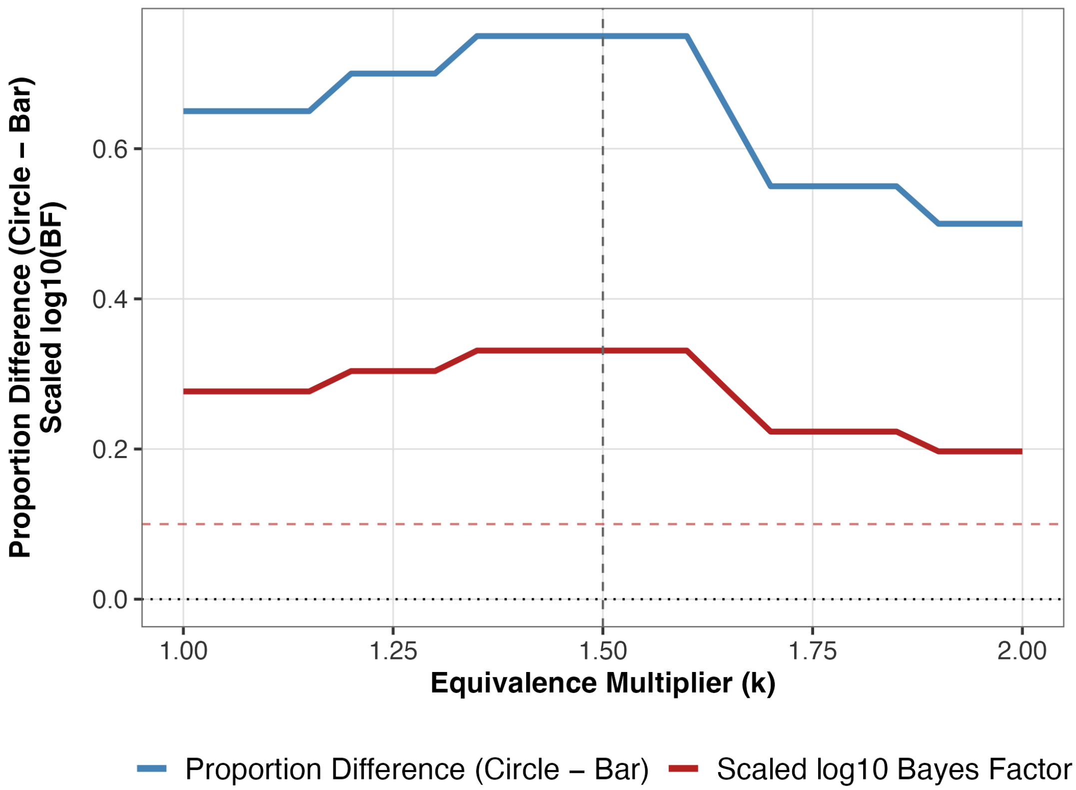

Figure 9

Experiment 2 (H2): sensitivity of the structural effect to the equivalence multiplier k*. Solid steelblue: proportion difference (circle minus bar) in the prevalence of joint target–and–decoy inclusion ($A \wedge S$). Solid firebrick: scaled log*$_{10}$ *Bayes factor,* $\log_{10}(\mathrm{BF}_{10})/10$*, from the paired-binary comparison at each* k*. Dashed grey vertical line: preregistered* $k = 1.5$*. Dotted black horizontal line at* 0*: null proportion difference. Dashed firebrick horizontal line at* 0.1*: corresponds to* $\mathrm{BF}_{10} = 10$ *on the scaled axis (*$\log_{10}(10)/10$*). Numeric ranges from this run (*$N = 20$ *participants): proportion difference* 0.542–0.792*;* BF_{10} 585.14–26214.40.

S2: Illustration of equivalence band estimation (Expe 2)

To clarify the construction and interpretation of the equivalence dispersion metric (VE_{rel}), we illustrate how participant-level equivalence bands are derived from repeated adjustment responses.

In the adjustment task, participants repeatedly adjusted one attribute of an option until it was perceived as equivalent to a fixed reference under the task rule (total fill for bars; value-for-money for circles). Each adjustment yields a response that can be expressed in a common task-relevant value scale (f), defined as $f = h + v$ for bars and $f = \mathrm{price}/\mathrm{area}$ for circles. For each trial, we computed the relative error:

$$e_{\mathrm{rel}} = \frac{f_B - f_A}{f_A},$$

where f_A and f_B denote the criterion values of the reference and adjusted options, respectively. The dispersion of these relative errors across repeated trials defines the participant's equivalence variability:

$$VE_{\mathrm{rel}} = \mathrm{SD}(e_{\mathrm{rel}}).$$

Supplementary Figure 10 visualizes this construction. Each point corresponds to a single adjustment response expressed in relative error units. The shaded region denotes the participant-specific equivalence band defined as $\pm k \cdot VE_{\mathrm{rel}}$ (with $k = 1.5$ in the preregistered analysis). Responses falling within this band are interpreted as lying within the participant's subjective equivalence region.

Although adjustments are made along individual attributes (e.g., horizontal vs. vertical extent, or price vs. area), all responses are expressed in a common value scale prior to aggregation. This transformation projects adjustment variability from the underlying attribute space onto deviations from the equivalence manifold (i.e., the set of value-matched configurations). As a result, the equivalence band reflects dispersion in value space rather than along any single physical dimension.

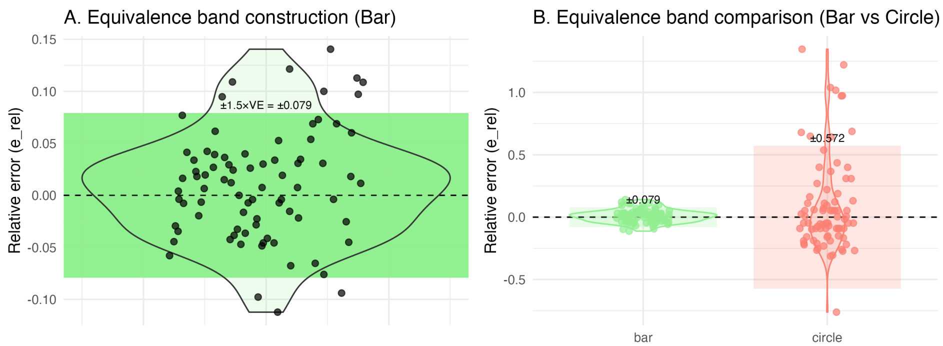

Figure 10

Illustration of equivalence band estimation in Experiment 2. Each point represents a single adjustment response expressed as relative error (e_{rel}) with respect to the reference criterion. The shaded region denotes the participant-specific equivalence band defined as $\pm k \cdot VE_{\mathrm{rel}}$ ($k = 1.5$). Panel A shows the construction of the equivalence band from trial-level variability for a representative participant. Panel B compares equivalence bands across stimulus types (bar vs. circle) for the same participant, illustrating greater dispersion under circle stimuli.

All equivalence bands are computed at the participant level; the figure illustrates a representative participant for visualization purposes. Panel A illustrates the construction of the equivalence band from trial-level variability. Panel B compares equivalence bands across stimulus types for the same participant. Consistent with the main analysis, circle stimuli exhibit greater dispersion (larger VE_{rel}) than bar stimuli, resulting in a broader equivalence region.

Notably, larger VE_{rel} implies a wider equivalence band, increasing the likelihood that externally defined offsets (target and decoy) fall within the subjective equivalence region. This band therefore serves as the key link between individual-level variability in equivalence judgments and the structural (gate-based) analysis of target and decoy inclusion.

S3: Continuous distance analysis (robustness check; Expe 2)

To assess whether the structural effect depends on the binary inclusion criterion, we conducted a complementary analysis using a continuous measure of proximity to the equivalence boundary. Instead of classifying offsets as inside or outside a band defined by $\pm k \cdot VE_{\text{rel}}$, we quantified the standardized distance of each offset from the boundary.

For each participant and stimulus type, the target and decoy offsets (Δ_T and Δ_D) were expressed in units of the participant's equivalence dispersion as:

$$d = \frac{|\Delta|}{VE_{\text{rel}}},$$

where $d = 1$ corresponds to the equivalence boundary under $k = 1$. Values $d < 1$ indicate that the offset lies within the participant's equivalence region, whereas $d > 1$ indicates that it lies outside.

Supplementary Figure 11 plots participant-level standardized decoy distances for bar (x-axis) and circle (y-axis) stimuli. Each point represents one participant. The diagonal line indicates equality between conditions; points falling below the diagonal indicate that the decoy lies closer to the equivalence boundary under circle stimuli than under bar stimuli. Dashed reference lines at $d = 1$ denote the equivalence boundary.

The distribution shows a systematic shift below the diagonal, indicating that decoy offsets are closer to the equivalence boundary for circle stimuli relative to bar stimuli. This pattern mirrors the binary gate analysis and demonstrates that the observed structural asymmetry does not depend on the discretization imposed by the equivalence band.

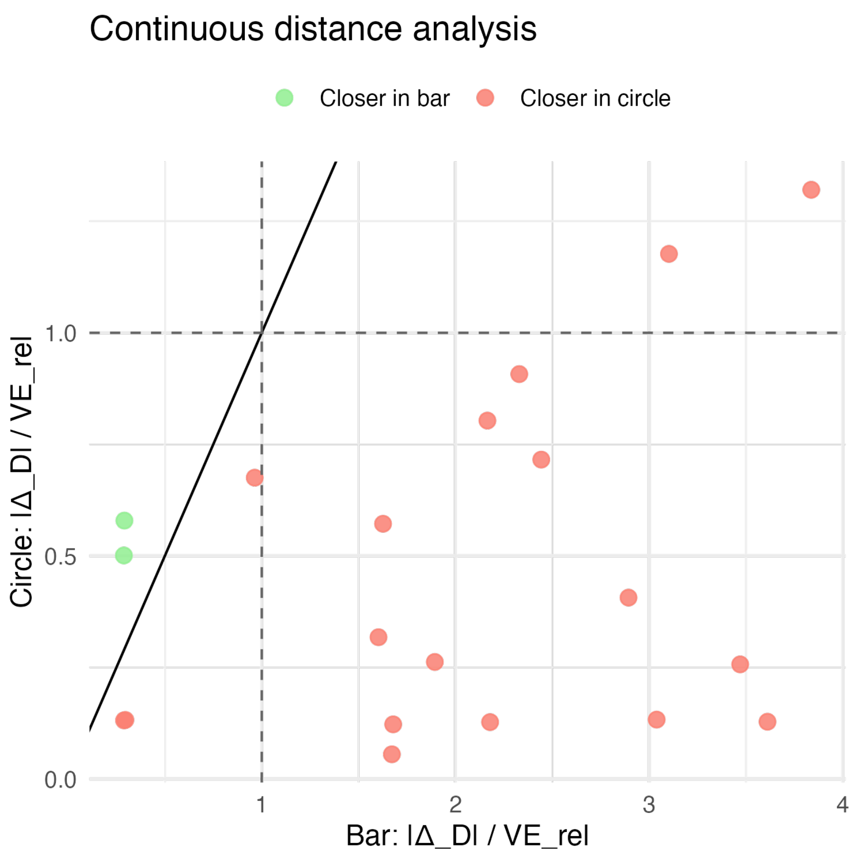

Figure 11

Continuous robustness analysis for Experiment 2. Each point represents a participant's standardized decoy distance ($|\Delta_D|/VE_{\mathrm{rel}}$) for bar (x-axis) and circle (y-axis) stimuli. The diagonal line indicates equality between conditions. Points below the diagonal indicate that the decoy lies closer to the equivalence boundary under circle stimuli. Dashed lines at $d = 1$ denote the equivalence boundary.

S4: Bayes Factor Trajectories (Exp 2)

To assess evidence accumulation over sample size, we computed sequential Bayes factors for both preregistered hypotheses using prefix samples, incrementally adding participants from $n = 3$ up to the final sample size (N). At each step, Bayes factors were recomputed using all available participants up to that point.

For H1 (dispersion), Bayes factors were computed from the paired-difference model comparing VE_{circle} and VE_{bar}, using the preregistered Cauchy prior on standardized effect size. For H2 (structural ambiguity), Bayes factors were computed using a paired binary beta–binomial model applied to discordant gate classifications (circle-only vs. bar-only), consistent with the confirmatory analysis pipeline.

Supplementary Figures 12 and 13 show the evolution of evidence as a function of sample size. In both cases, evidence increasingly favors the directional alternatives as participants accumulate, converging with the final Bayes factor estimates reported in the main text.

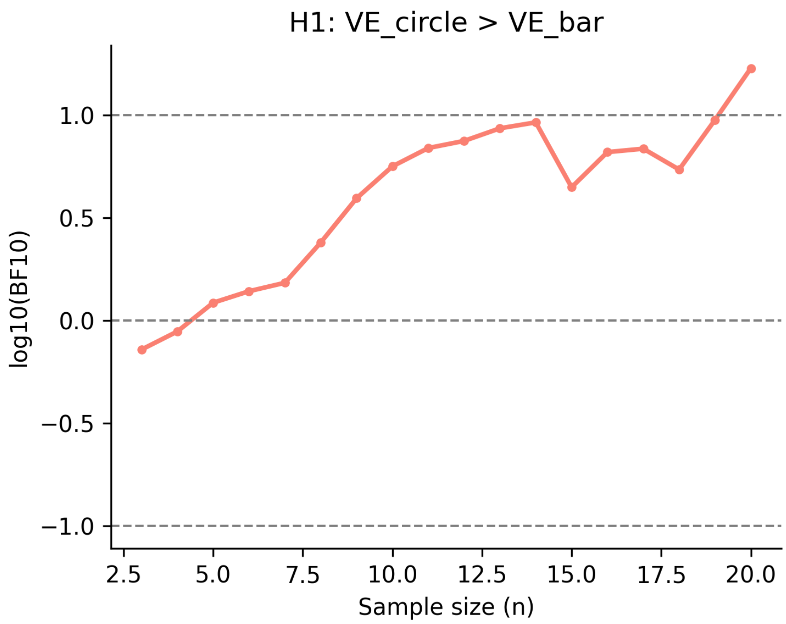

Figure 12

Sequential Bayes factor trajectory for H1 in Experiment 2. Bayes factors (BF_{10}) for the hypothesis $VE_{\mathrm{circle}} > VE_{\mathrm{bar}}$ are shown as a function of cumulative sample size. Each point reflects the Bayes factor computed from all participants up to that sample size.

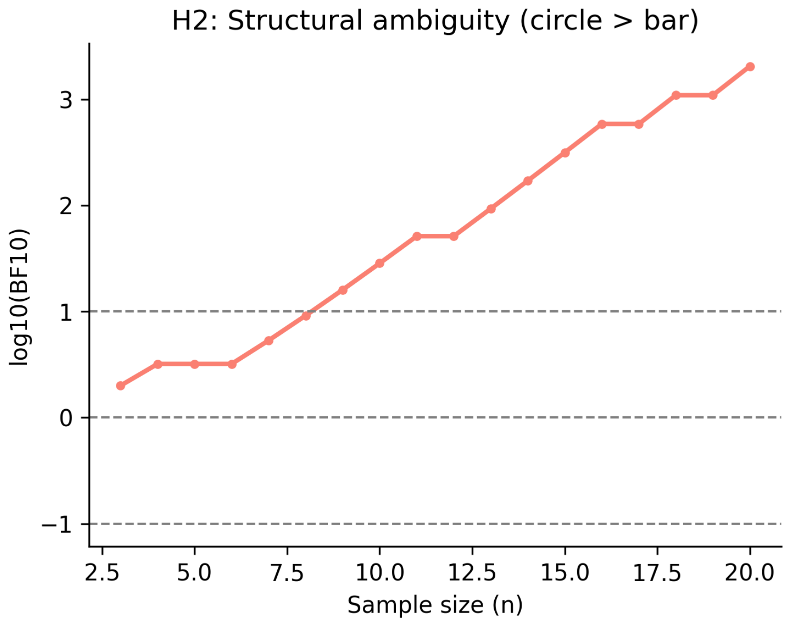

Figure 13

Sequential Bayes factor trajectory for H2 in Experiment 2. Bayes factors (BF_{10}*) for the hypothesis of greater joint structural ambiguity under circle relative to bar stimuli are shown as a function of cumulative sample size. Each point reflects the Bayes factor computed from discordant gate classifications across all participants up to that sample size.*

S5. Cognitive Foundations and Fraction Stimulus Space

Fractions provide an ideal stimulus class for precisely manipulating pairwise comparison difficulty because their evaluation is heavily influenced by the *natural number bias* (Alibali & Sidney, 2015; Ni & Zhou, 2005; Reinhold et al., 2023). While humans can process whole numbers rapidly (Moyer & Landauer, 1967), the automatic processing of these symbolic numbers frequently interferes with judgments of a fraction's holistic magnitude (Lortie-Forgues et al., 2015).

Specifically, decision-makers often mistakenly rely on the independent magnitudes of the numerator and denominator rather than their integrated ratio. For example, a fraction comprising larger numbers (e.g., $\frac{1283}{11823}$) is often incorrectly judged as larger than a fraction with smaller numbers (e.g., $\frac{1}{3}$) because of the automatic interference of whole-number processing (Behr et al., 1984). Conversely, studies have also identified a *reverse natural number bias*, where individuals judge fractions with smaller numbers as the larger fractions—a likely overcorrection to the standard natural number bias (Barraza et al., 2017; DeWolf & Vosniadou, 2011; DeWolf & Vosniadou, 2015; Obersteiner et al., 2020).

To accurately compare fractions and navigate these opposing biases, decision-makers must suppress the urge to solely compare number magnitudes. Instead, they rely on a complex mix of componential strategies (e.g., comparing gaps between numerators and denominators) and holistic heuristics (e.g., benchmarking to familiar fractions) (Obersteiner et al., 2022; Siegler et al., 2013).

To systematically manipulate these comparative difficulties, all fraction stimuli in the present experiment were drawn from the space of rational numbers with integer numerators and denominators greater than 1 and less than 100, excluding cases in which the numerator equaled the denominator. This space was chosen to avoid trivial equivalences while allowing sufficient flexibility to construct controlled triplets that either align with or bypass the cognitive bottlenecks described above. Potential fraction triplets were sampled randomly from this space and evaluated against the specific heuristic selection criteria detailed below.

S6. Ease of Holistic Magnitude Perception

To construct the stimulus set, we first needed to operationalize the fluency with which a decision-maker can perceive the holistic magnitude (i.e., the integrated "super-value") of an individual fraction. The symbolic notation of fractions often obscures their true magnitude (Lortie-Forgues et al., 2015); however, specific numerical properties allow individuals to process fractions more holistically and bypass the natural number bias (Park et al., 2021).

We quantified the ease of holistic perception for each individual fraction using a geometric mean (with a 0.05 offset) of four binary criteria, grounded in the fraction cognition literature:

1. **Perfect Divisibility:** The numerator is perfectly divisible by the denominator. Fractions that evaluate to whole numbers are processed significantly faster and more accurately because they map directly onto existing whole-number representations (DeWolf & Vosniadou, 2011; Obersteiner et al., 2022).

2. **Whole-Number Proximity:** The ratio represented by the fraction is within ± 0.15 of a whole number. This facilitates rapid "benchmarking," a prevalent strategy where decision-makers round fractions to the nearest familiar integer to estimate magnitude (Liu, 2018).

3. **High Simplifiability:** The numerator and denominator share 4 or more common factors. High commonality allows for rapid mental simplification, which reduces the working-memory load required to integrate the two components (Fazio et al., 2016; Reinhold et al., 2023).

4. **Base-10 Compatibility:** The denominator is divisible by 5. Denominators that are

factors of base-10 systems (e.g., 5, 10) facilitate rapid translation into familiar decimal equivalents, heavily reducing evaluation difficulty (Siegler et al., 2013).

S7. Pairwise Ease-of-Comparison Heuristics

To guide stimulus selection for pairwise comparisons, we defined an ease-of-comparison score for each fraction pair. Rather than relying on arbitrary mathematical distances, we mathematically formalized several well-documented cognitive strategies used in fraction processing (Obersteiner et al., 2022; Siegler et al., 2013). Evaluating fractions is notoriously difficult because the automatic processing of symbolic natural numbers often interferes with judgments of overall magnitude—a phenomenon known as the *natural number bias* (Alibali & Sidney, 2015; Ni & Zhou, 2005; Reinhold et al., 2023). To navigate this, decision-makers rely on a mix of componential and holistic heuristics, which we integrated into a geometric-mean metric to define comparative ease.

The pairwise ease score was calculated as the geometric mean of a series of binary heuristic conditions. To prevent the geometric mean from evaluating to zero, the presence of a condition was encoded as 1.05 and its absence as 0.05. These conditions were directly informed by the literature on fraction comparison strategies:

1. **Common Components:** The fractions share a common component (numerator or denominator). Because this bypasses complex integration and allows for direct magnitude comparison, this condition was weighted four times more heavily than the others in the geometric mean calculation.

2. **Gap Congruence:** Decision-makers frequently employ a componential strategy of comparing the differences between numerators and denominators across fractions (Gómez & Dartnell, 2019; Obersteiner et al., 2022). This condition was met if the comparison was gap congruent (i.e., the item with the larger

numerator-denominator difference is the objectively larger fraction) and the ratio between the gaps of the two fractions was ≥ 1.5.

3. **Numerator Benchmarking:** The ratio between the numerators was ≤ 5.05 (capturing the most familiar portions of the multiplication table) and was within ± 0.15 of a whole number. This aligns with decision-makers' reliance on familiar whole-number benchmarks (DeWolf & Vosniadou, 2015; Liu, 2018).

4. **Denominator Benchmarking:** The ratio between the denominators met the exact same criteria as the numerators (≤ 5.05 and within ± 0.15 of a whole number).

5. **Holistic Magnitude Distance:** The strength of the natural number bias decreases when the holistic distance between two fractions is large (Barraza et al., 2017; Park et al., 2021). This condition was met if the ratio between the holistic magnitudes of the fractions was either > 2 (very distant) or within 1 ± 0.15 (very close).

These heuristics were used strictly to constrain stimulus selection and ensure a controlled manipulation of pairwise comparative asymmetry (ΔDiff); they do not constitute claims about the explicit strategies participants employed on any given trial.

The composite ease-of-comparison score for each pairwise comparison was computed as the geometric mean of these conditions, with the Common Components condition weighted four times more heavily. For each trial, three pairwise ease values were calculated: Target–Decoy (Ease$_{TD}$), Competitor–Decoy (Ease$_{CD}$), and Target–Competitor (Ease$_{TC}$). The comparative asymmetry manipulation (ΔDiff) was operationalized as:

$$\Delta\text{Diff} = \text{Ease}_{TD} - \text{Ease}_{CD}.$$

Figure 14 displays the mean ease scores across the three pairwise comparisons for the High Diff and Low Diff conditions. In the High Diff condition, Target–Decoy comparisons were substantially easier ($M = 0.458$, $SD = 0.097$) than Competitor–Decoy comparisons ($M = 0.081$, $SD = 0.034$), yielding ΔDiff $= 0.377$. The Target–Competitor comparison remained approximately constant ($M = 0.169$, $SD = 0.056$). In the Low Diff condition, the asymmetry was substantially reduced: Ease$_{TD}$ ($M = 0.748$, $SD = 0.078$) and Ease$_{CD}$ ($M = 0.571$, $SD = 0.089$) were more similar, yielding ΔDiff $= 0.177$, while Ease$_{TC}$ remained constant ($M = 0.169$, $SD = 0.049$). This manipulation successfully isolated decoy-related comparative asymmetry while holding core-option difficulty (Target–Competitor ease) approximately constant.

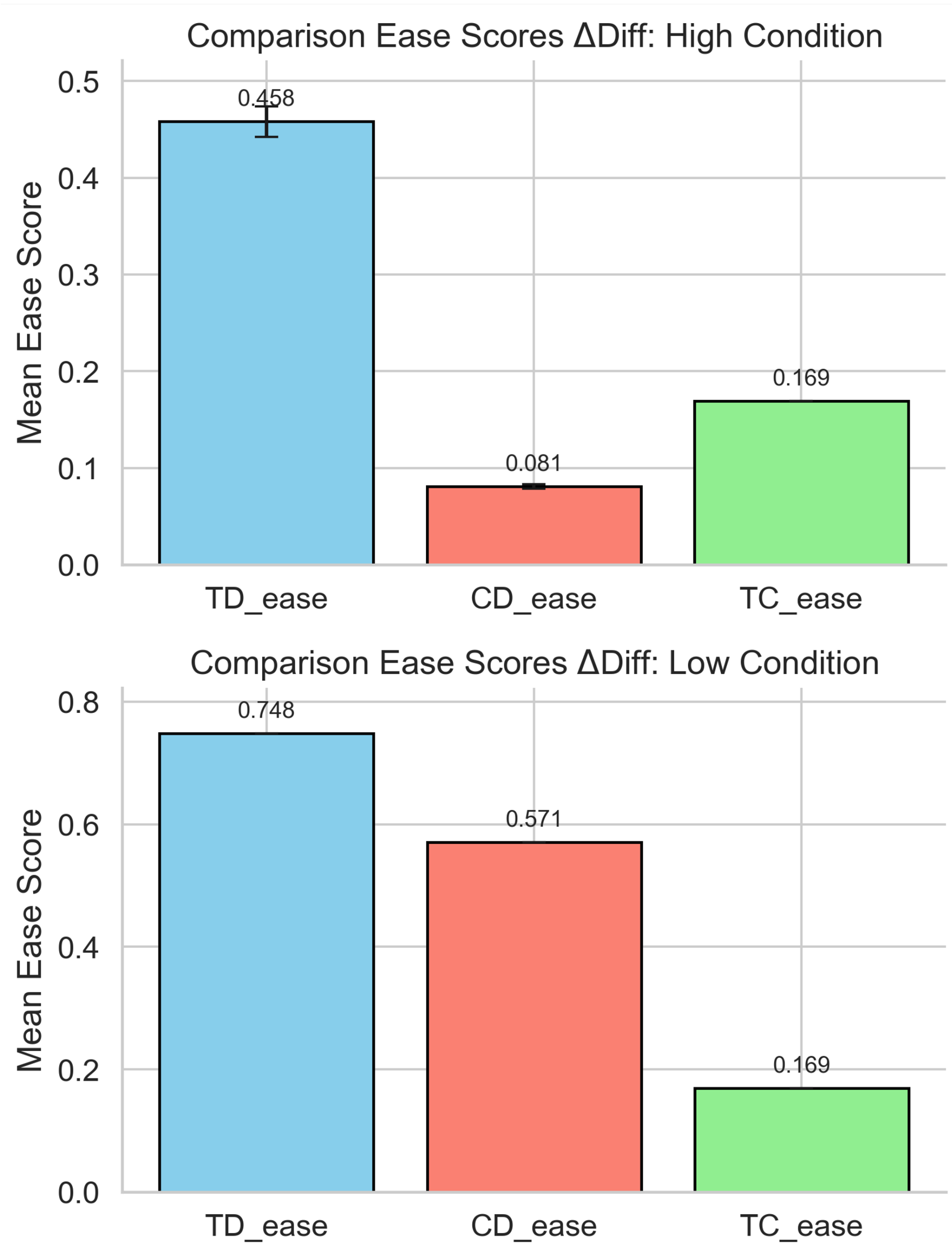

Figure 14

Mean Pairwise Ease-of-Comparison Scores across High Diff and Low Diff Conditions. Error bars represent standard deviations. TD_{ease} *= Target–Decoy ease;* CD_{ease} *= Competitor–Decoy ease;* TC_{ease} *= Target–Competitor ease. High Diff condition (*$\Delta Diff = 0.377$*) shows substantial asymmetry favoring Target–Decoy comparisons. Low Diff condition (*$\Delta Diff = 0.177$*) shows reduced asymmetry. Target–Competitor ease remains approximately constant across both conditions, validating the core-option ambiguity control.*

S8. Final Stimulus Set and Triplet Construction

Using the pairwise ease scores defined above, three comparisons were evaluated for each candidate triplet: Target–Decoy (TD), Competitor–Decoy (CD), and Target–Competitor (TC).

In the *Low* $\triangle$Diff condition, the ratio between TD and CD ease scores was constrained to be < 2, and a minimum ease threshold of 0.4 was imposed for both comparisons. In the *High* $\triangle$Diff condition, the ratio between TD and CD ease scores was constrained to be > 4, ensuring the decoy was substantially easier to compare with the target than with the competitor. Across both conditions, TC ease scores were constrained to lie within a fixed range (ratios between 2 and 5.5) and were further restricted to avoid values within ± 0.15 of a whole-number ratio. These constraints ensured that Target–Competitor comparison difficulty remained approximately constant across conditions.

Following the application of all constraints, 30 base fraction triplets were selected. Each triplet was instantiated in two contextual variants corresponding to the two decoy constructions, yielding 60 trials per condition. Crucially, the exact same Target–Competitor pairs appeared in both conditions, differing only in the relative construction of the decoy.

S9: Pair-Rating Validation of the $\triangle$Diff Manipulation

This supplementary analysis validates whether the fraction-stimulus manipulation used in Experiment 3 was psychologically realized at the level of pairwise comparative difficulty. Specifically, we tested whether perceived difficulty differences between Competitor–Decoy (CD) and Target–Decoy (TD) comparisons were larger in the High $\triangle$Diff condition than in the Low $\triangle$Diff condition, while Target–Competitor difficulty remained stable across conditions.

Task and Design

Participants completed a pair-rating task in which they made pairwise choices and then rated subjective difficulty on a 1–7 scale. The design included two within-participant blocks corresponding to the same two comparative-asymmetry conditions used in Experiment 3: Low $\triangle$Diff and High $\triangle$Diff. In the log files, the three pair types were coded as `TD`, `CD`, and `CT`; `CT` corresponds to Target–Competitor (TC) in the manuscript notation.

The initial sample was $N = 58$ participants. In this supplementary analysis, we applied catch-based exclusion only (10 participants), yielding a final retained sample of $N = 48$. Due to a technical logging issue, 17 participants had fewer RT rows than expected ($< 80\%$ of 110 expected trials); however, we did not exclude participants on that basis here. Results were qualitatively similar with and without that missingness exclusion.

Descriptive Pairwise Difficulty Patterns

Figure 15 and Figure 16 show participant-level mean difficulty ratings by pair type in each condition. The same directional pattern was observed in both blocks: CD comparisons were rated as more difficult than TD comparisons. The absolute CD–TD separation was visibly larger in the High $\triangle$Diff block.

Inferential Results

Primary manipulation-check contrast: change in the CD–TD gap. Critically, the participant-level CD–TD difficulty gap was larger in the High $\triangle$Diff block ($M = 1.01, SD = 0.72$) than in the Low $\triangle$Diff block ($M = 0.27, SD = 0.29$). This increase was reliable: paired t-test $t(47) = 7.84, p < .001, d_z = 1.13$, mean difference (High–Low) $= 0.73$, 95% CI $[0.55, 0.92]$. A Wilcoxon signed-rank robustness test yielded the same conclusion, $W = 41.50, p < .001$.

Target–Competitor (TC) stability check. To test whether core-option comparison difficulty remained stable, we compared `CT` ratings (log code for TC) across blocks. Mean CT difficulty was $M = 3.44$ ($SD = 0.92$) in Low $\triangle$Diff and $M = 3.44$

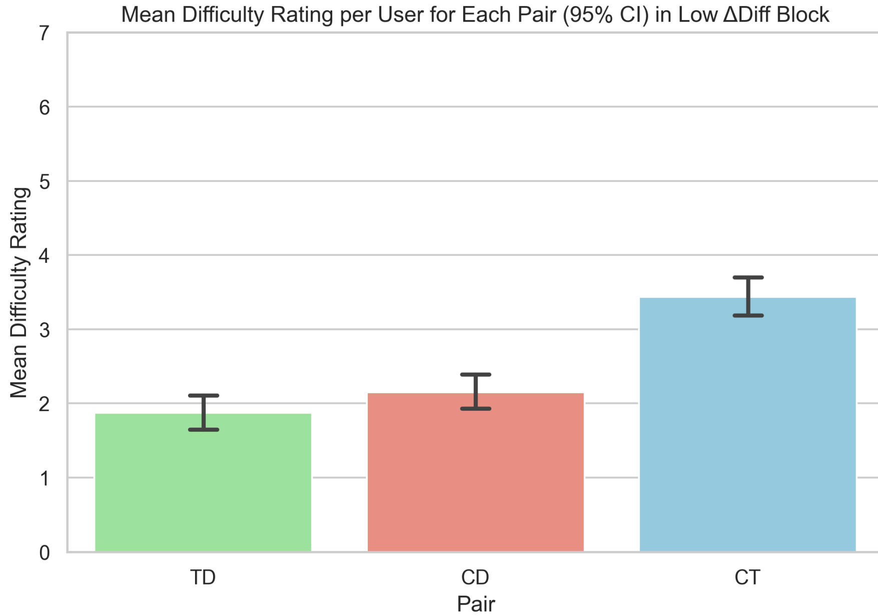

Figure 15

Mean difficulty rating per participant for each pair type in the Low △Diff block. Error bars indicate 95% confidence intervals across participant means.

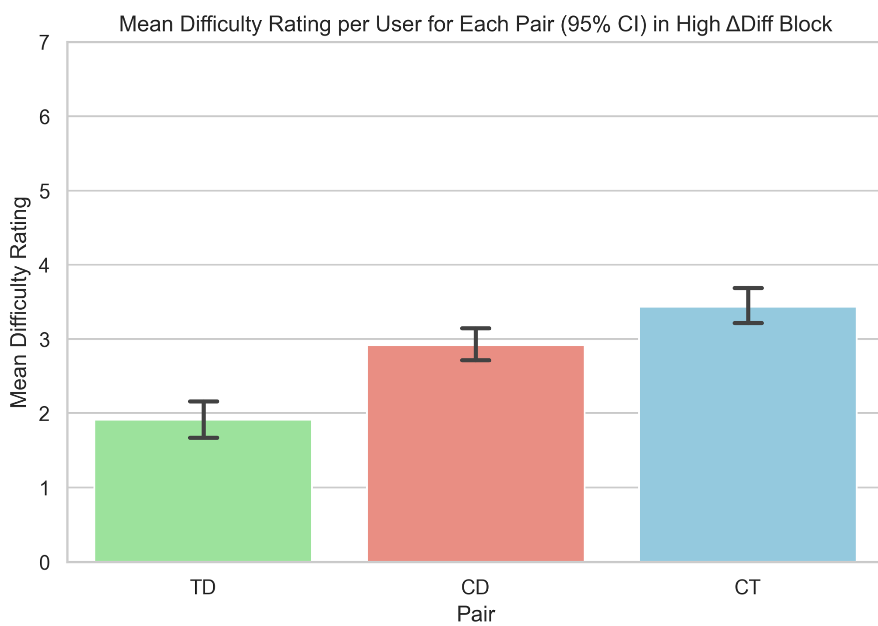

Figure 16

Mean difficulty rating per participant for each pair type in the High △Diff block. Error bars indicate 95% confidence intervals across participant means.

($SD = 0.84$) in High ΔDiff. This difference was not reliable: paired t-test $t(47) = -0.03, p = .977, d_z = -0.00$, mean difference $= -0.00$, 95% CI $[-0.11, 0.11]$; Wilcoxon signed-rank test $W = 513.00, p = .959$.

Interpretation

Taken together, these supplementary results support the intended manipulation structure for Experiment 3. The decoy-related asymmetry signature (CD harder than TD) is substantially amplified in High ΔDiff, while Target–Competitor (TC) difficulty remains statistically unchanged across blocks. This pattern is consistent with selectively increasing decoy-related comparative asymmetry rather than globally shifting all pairwise comparison difficulty.

S10. Overview and Modeling Rationale

Experiment 3 aimed to isolate the role of asymmetric comparative processing in producing attraction effects while controlling for Target–Competitor comparison difficulty. The core theoretical claim is that attraction reflects a two-stage process: first, the selective elimination of the decoy from consideration, driven by the comparative ease of the Target–Decoy relation; and second, the formation of relative preference between the target and competitor once the decoy has been removed. Rather than treating attraction as a unitary shift in target-choice frequency, this decomposition reflects the theoretical distinction between rejecting dominated alternatives and forming preferences among non-dominated ones (Katsimpokis et al., 2022; Liew et al., 2016).

To implement this decomposition, we analyzed choice behavior at the trial level using two separate hierarchical Bayesian logistic regression models. Hierarchical Bayesian logistic regression provides a natural framework for this analysis by simultaneously accommodating trial-level predictors, individual differences via participant-specific random effects, and uncertainty in parameter estimates. The first model modeled the probability of decoy choice across all trials, with log-transformed response time (RT), ΔDiff condition, and block order as predictors. This model assessed whether extended deliberation selectively reduced decoy choice and whether that reduction depended on the comparative asymmetry induced by the manipulation. The second model was restricted to trials on which the decoy was not selected, and modeled target choice as a function of the same predictors.

Both models were estimated in a Bayesian framework using weakly informative priors, which regularize estimates without imposing strong theoretical assumptions about the magnitude or direction of effects. Posterior inference relied on credible intervals and Region-of-Practical-Equivalence (ROPE) analyses rather than null-hypothesis significance testing, allowing direct assessment of whether observed effects are practically meaningful rather than merely statistically detectable (Kruschke & Liddell, 2018). Model comparisons and robustness checks—including Pareto-smoothed leave-one-out cross-validation (PSIS-LOO) (Vehtari et al., 2017)—were included to demonstrate that the substantive conclusions do not depend on specific modeling choices. Full specifications of priors, ROPE bounds, and model-fit diagnostics are provided in S11–S14 below.

S11. Decoy Choice Model

Decoy choice was modeled using a hierarchical logistic regression applied to all trials. Let D_{ij} denote whether participant j selected the decoy on trial i ($D_{ij} = 1$ if the decoy was chosen, 0 otherwise). The model was specified as:

$$D_{ij} \sim \text{Bernoulli}(\pi_{ij}),$$

$$\text{logit}(\pi_{ij}) = \beta_0 + \beta_1 \log(\text{RT}_{ij}) + \beta_2 \Delta\text{Diff}_{ij} + \beta_3 \text{BlockOrder}_j + \beta_4 \log(\text{RT}_{ij}) \times \Delta\text{Diff}_{ij} + u_{0j} + u_{1j} \log(\text{RT}_{ij}),$$

where $\log(\text{RT}_{ij})$ is the log-transformed response time, ΔDiff_{ij} denotes the comparison-difficulty manipulation (high vs. low), and BlockOrder_j indicates block order (HL vs. LH). Participant-specific random intercepts (u_{0j}) and random slopes for logRT

(u_{1j}) captured individual differences in baseline decoy avoidance and time sensitivity.

This model assessed whether extended deliberation selectively reduced decoy choice and whether this reduction depended on the asymmetric comparison structure induced by the ΔDiff manipulation.

S12. Target Choice Conditional on Decoy Elimination

To assess whether attraction effects reflect changes in target–competitor preference beyond decoy elimination, target choice was analyzed conditional on trials in which the decoy was not selected. Let T_{ij} denote whether participant j chose the target on trial i, given $D_{ij} = 0$.

$$T_{ij} \mid (D_{ij} = 0) \sim \text{Bernoulli}(\theta_{ij}),$$

$$\text{logit}(\theta_{ij}) = \gamma_0 + \gamma_1 \log(\text{RT}_{ij}) + \gamma_2 \Delta\text{Diff}_{ij} + \gamma_3 \text{BlockOrder}_j + \gamma_4 \log(\text{RT}_{ij}) \times \Delta\text{Diff}_{ij} + v_{0j} + v_{1j} \log(\text{RT}_{ij}).$$

This conditional analysis isolates target–competitor preference formation after the decoy has been eliminated, allowing direct assessment of whether response time or comparison structure shifts relative preference rather than merely suppressing decoy choice.

S13. Prior Specification

All priors were chosen to be weakly informative, providing regularization without favoring either the presence or absence of attraction effects.

Fixed-effect coefficients were assigned normal priors:

$$\beta_k, \gamma_k \sim \mathcal{N}(0, 1),$$

corresponding to modest prior beliefs about plausible log-odds effects.

Standard deviations of participant-level random effects were assigned half-Student-t priors:

$$\sigma \sim \text{Half-Student-}t(3, 0, 1),$$

and correlation matrices were assigned LKJ priors:

$$\mathbf{R} \sim \text{LKJ}(2).$$

These priors regularize extreme individual differences while allowing substantial heterogeneity across participants.

S14. Region of Practical Equivalence (ROPE)

To assess evidence for practically meaningful effects, posterior distributions were evaluated using ROPE analyses (Kruschke & Liddell, 2018). For regression coefficients expressed in log-odds units, the ROPE was defined as $[-0.1, 0.1]$, corresponding approximately to differences in choice probability smaller than 2–3% across the observed range of predictors.

Posterior mass falling largely within the ROPE was interpreted as evidence for a practically negligible effect, whereas posterior mass predominantly outside the ROPE was interpreted as evidence for a meaningful effect. Intermediate cases were classified as inconclusive.

S15. Model Robustness and Predictive Adequacy

As a robustness check, we evaluated the predictive adequacy of all hierarchical models using Pareto-smoothed importance sampling leave-one-out cross-validation (PSIS-LOO; (Vehtari et al., 2017)). For both the decoy-elimination model and the target-choice model, all Pareto-k diagnostics were below 0.7, indicating that posterior estimates were not driven by influential observations and that the LOO approximation was reliable. The effective number of parameters (p_{loo}) was consistent with the hierarchical structure of the models. Monte Carlo standard errors of the expected log predictive density were negligible, confirming stable posterior inference. These results indicate that the reported effects are robust and generalize beyond the observed data.